UTILIZATION OF DEEP LEARNING: OPTIMIZED YOLOV8 FOR DETECTION OF MOTORCYCLE INTRUSION AND LICENSE PLATE RECOGNITION ON BICYCLE LANES

A Thesis Presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna

In partial fulfillment of the requirement for the degree
Bachelor of Science in Computer Science
Specialized in Intelligent Systems

DAREL T. ALCARAZ
CHRIS F. BALOGO
RODA C. MENDEZ

JEFFERSON L. LERIOS

June 2026

APPROVAL SHEET

The thesis entitled **"UTILIZATION OF DEEP LEARNING: OPTIMIZED YOLOV8 FOR DETECTION OF MOTORCYCLE INTRUSION AND LICENSE PLATE RECOGNITION ON BICYCLE LANES"** prepared and submitted by **Darel T. Alcaraz, Chris F. Balogo, and Roda C. Mendez** in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science specialized in Intelligent Systems, is hereby recommended for approval and acceptance.

JEFFERSON L. LERIOS
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of ____.

WILSON B. SUYAT
Member

CAROLINA R. JOVAL
Member

JOCELYN O. PADALLAN
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science specialized in Intelligent Systems.

LOYD S. ECHALAR, DIT
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

ACKNOWLEDGEMENT

This thesis would not be possible without the invaluable contributions, insights, and support of the several individuals who played key roles in the completion of this work. It is a pleasure to give gratitude to those who made it possible.

The researchers sincerely thank the University President, Dr. Mario R. Briones, and the OIC Campus Director, Engr. Manuel Luis R. Alvarez, for their excellent leadership and conducive academic environment they foster on our university. To our beloved College of Computer Studies Associate Dean, Dr. Loyd S. Echalar, for his warm advises and consistent encouragement.

Our appreciation also goes to Prof. Jefferson L. Lerios, our thesis adviser, for his invaluable guidance, unwavering support, and insightful contribution throughout our journey with this project. Simultaneously, we would like to thank Mr. Wilson B. Suyat for his invaluable expertise and unwavering guidance as our subject specialist, Ms. Carolina R. Joval for her meticulous statistical analysis and insights, and Ms. Jocelyn O. Padallan for ensuring our paper obeyed to the required technical format. Additionally, we would like to extend our gratitude to Mr. Gene Marck B. Catedrilla and Mr. Jonardo R. Asor, and the rest of our prefessors for their endless help and support, which greatly enriched our thesis process.

Finally, we thank Almighty God, for his provision of our needs, knowledge, perseverance, and guidance to accomplish this research.

The Researchers

DEDICATION

The success of this project is respectfully and wholeheartedly dedicated by the researchers to our Almighty God, who gave them knowledge, protection, and guidance. To their families:

Maribel T. Alcaraz and Dante C. Alcaraz

Arlene F. Balogo and Crisanto A. Balogo

Arlene C. Mendez and Rodrigo T. Mendez

For their boundless love, unwavering support, and immense sacrifices, both financial and moral, were the foundation of their perseverance and dedication to work hard to the project.

To their mentors whose insightful guidance, profound wisdom, and continuous encouragement were invaluable at every stage of this study. Their expertise significantly shaped the development and successful completion of this study.

To their friends, partners, and siblings, who provided encouragement during times of hardship and difficulty, reminding them to persist.

D.T.A.

C.F.B.

R.C.M.

ABSTRACT

The study utilized a deep learning algorithm, specifically optimizing the YOLOv8 to detect motorcycle intrusions and recognize license plates using EasyOCR on bicycle lanes. As motorcycle usage rises in urban areas, the need for effective monitoring systems becomes critical for protecting cyclists and pedestrians. The research involves collecting and annotating a dataset of 1,832 images using Roboflow, leading to the development of two models: YOLOv8 for vehicle and license plate detection and U-Net with ResNet34 for bicycle lane segmentation. Model performance was assessed using metrics like Precision, Recall, F1 Score, and Mean Average Precision (mAP) for detection, and Dice coefficient and Intersection over Union (IoU) for segmentation accuracy. EasyOCR was also integrated for license plate recognition, evaluated through Character Recognition Rate (CRR) and Plate Recognition Rate (PRR). Results indicated that the YOLOv8 model achieved 88% precision and 91% mAP while the segmentation model had an IoU of approximately 0.975, and EasyOCR demonstrated a character accuracy of 83.82%. The YOLOv8 model detected motorcycles on bicycle lanes with high accuracy, resulting in a precision of 94.74%, recall of 87.54%, and an F1-score of 91%. Further, statistical test results showed no significant relationship between the expert count and the model count, with a p-value of 0.594. This results show that the model's performance is closely aligned with the experts' assessment, supporting the reliability of YOLOv8 model in detection of motorcycle intrusion on bicycle lanes.

Keywords: *You Only Look Once Version 8 (YOLOv8), Motorcycle, EasyOCR, Bicycle Lanes, U-Net, ResNet34, Detection, Annotation, Recognition, Deep Learning*

TABLE OF CONTENTS

Preliminaries	Page
Title Page	i
Approval Sheet	ii
Acknowledgement	iii
Dedication	iv
Abstract	v
Table of Contents	vi
List of Figures	viii
List of Tables	x
Definition of Terms	xi

CHAPTER

1	**INTRODUCTION**	
	Background of the Study	1
	Objectives of the Study	5
	Significance of the Study	6
	Scope and Limitations	7
2	**REVIEW OF RELATED LITERATURE AND STUDIES**	
	Related Literature	9
	Literature Matrix	21
	Synthesis and Relevance of the Study	23
	Conceptual Framework	24
3	**METHODOLOGY**	
	Study Area	25
	Materials	26
	Software	26
	Hardware	29
	Research Design	29
	Model Development	30
	Data Acquisition	32
	Model Selection	34
	Model Evaluation	36
	Model Validation	38
	Ethical Consideration	40
4	**RESULTS AND DISCUSSIONS**	
	Gathered Data	42
	Model Development Results	43
	Object Detection	44
	Instance Segmentation	46
	License Plate Recognition	46
	Motorcycle Intrusion on Bicycle Lanes	47
	Model Performance Metrics	48
	YOLOv8	49

U-Net with RestNet34 53
EasyOCR 55
Model Validation 56
Vehicle and License Plate Detection 56
Bicycle Lane Segmentation 59
Motorcycle Intrusion Validation 59

5 SUMMARY, CONCLUSIONS AND RECOMMENDATIONS
Summary 63
Conclusions 64
Recommendations 65

REFERENCES
APPENDICES
Appendix A. Letter of Approval
Appendix B. Certificate of Acceptance
Appendix C. Relevant Source Code
Appendix D. Evaluation Tool
Appendix E. Sample Input / Output
Appendix F. User's Guide
Appendix G. Matrix For Revision
Appendix H. Grammarly & Turnitin Report
Appendix I. Curriculum Vitae
Appendix J. Publishable Paper (IEEE format)

LIST OF FIGURES

No.	Title	Page
1	Conceptual Framework	24
2	Location of the Study Area Used in the Model Development	25
3.1	Model Development for Detection of Motorcycle Intrusion	32
3.2	Model Development for License Plate Recognition	32
4	Model Architecture	36
5	Sample Data of (a) Raw Data, (b) Annotated Bicycle Lane, and (c) Annotated Vehicles and License	42
6	Trained Sample Result Using YOLOv8	45
7	Trained Sample Result on Bicycle Lane Segmentation	46
8	Applying Preprocessing in License Plate with (a) Image Denoising and (b) Grayscale Conversion	47
9	Result for Detection of Motorcycle Intrusion and LPR	48
10	F1-Confidence	50
11	Precision-Confidence	51
12	Precision-Recall	52
13	Recall-Confidence	52
14	Results in (a) Dice Coefficient and (b) Intersection over Union (IoU)	53
15	Validation Losses and Learning Rate During Training	56
16	Confusion Matrix Normalized	57

17 Confusion Matrix 58

18 Training and Validation Loss 59

LIST OF TABLES

No.	Title	Page
1	Matrix of Reviewed Related Literature	21
2	Desktop Specification	29
3	Dataset Table	33
4	Optimized Hyperparameters of YOLOv8	44
5	Detection Evaluation Results	49
6	Model Configuration	54
7	Performance Summary	54
8	EasyOCR Evaluation Results	55
9	Performance Metrics Results	60
10	Statistical Test Results	61
11	Test of Normality (Shapiro–Wilk)	61
12	Descriptive Summary	62

DEFINITION OF TERMS

This section provides definitions for the key terms used in this study. These definitions help clarify the concepts, methods, and elements involved in optimizing the YOLOv8 model for detecting motorcycle intrusions and license plate recognition on bicycle lanes.

Annotation	The process of labeling objects, regions, or features in images (e.g., bounding boxes, polygons) to train computer vision models.
Augmentation	The process of labeling objects, regions, or features in images (e.g., bounding boxes, polygons) to train computer vision models.
Bicycle Lane	A designated portion of the road marked for bicycle use, separate from motor vehicle traffic.
Deep Learning	A subset of machine learning that uses multi-layered neural networks to extract patterns from large datasets.
EasyOCR	Use deep learning to extract text from images in multiple languages
Google Colaboratory (Colab)	Cloud-based environment that provides free access to GPUs for training and testing machine learning models.
Image Normalization	Scaling pixel values to a common range to ensure consistency in model input.
Image Resizing	Adjusting image dimensions to match the input size required by a neural network.

Lane Segmentation	The process of identifying lane boundaries in images for tasks like traffic monitoring.
Motorcycle Intrusion	The unauthorized entry of motorcycles into bicycle lanes or other restricted areas.
Object Detection	A computer vision task that identifies and localizes objects in images or videos.
Pre-processing	The set of techniques applied to raw images before training or inference, including resizing, normalization, filtering, and enhancement.
Region of Interest (ROI)	Selected area within an image targeted for focused analysis or detection.
ResNet / ResNet34	Deep neural network with skip connections that enable training of very deep models. ResNet34 is used in U- Net segmentation.
Roboflow	Web-based tool for dataset creation, annotation, augmentation, and preprocessing in computer vision projects.
U-Net	A deep learning architecture used for image segmentation, especially useful in biomedical and traffic-related applications.
YOLOV8s	A small and lightweight variant of YOLOv8 designed for high-speed performance with reduced computational requirements. Ideal for real-time detection on mobile or edge devices.

CHAPTER 1

INTRODUCTION

Background of the Study

The increasing use of motorcycles on city roads has been a serious concern, as motorists tend to violate traffic regulations by entering bicycle lanes. This act endangers not just pedestrians but also cyclists while compromising the designated safe zones intended for their use. The World Health Organization (2023) states that road traffic injuries result in an estimated 1.19 million deaths each year globally, with over half the cases involving vulnerable road users, including pedestrians (21%), cyclists (5%), and motorcyclists (30%). The WHO further states that 92% of road fatalities occur in low- and middle-income countries, despite having only 60% of the world's vehicles, highlighting significant gaps in traffic safety enforcement.

In the United States, the National Highway Traffic Safety Administration (NHTSA) reported 1,166 bicyclists were killed and an estimated 49,489 bicyclists were injured in traffic crashes in 2023. Additionally, pedestrian deaths accounted for 18% of all traffic fatalities in 2023 and 3% of all people injured in traffic crashes in 2023. These statistics show that people who do not use vehicles are more affected in traffic areas where there is not enough enforcement and monitoring.

In the Philippines, motorcycle accidents are one of the few problems the country is facing. Based on Statistica (2024), there were about 8.5 million registered motorcycles, tricycles, and non-conventional or three-wheeled motorcycles in the countries as of 2022. More so, there were at least 26,599 motorcycle accidents officially reported by the Road Safety Unit of the Metro-

politan-Manila Development Authority (MMDA, 2022). This number even increased by 17.3 percent in 2023. In simpler terms, there are at least 78 motorcycle accidents daily (Philstar, 2024). It shows that we need to enforce traffic laws more strictly, especially those that protect vulnerable road users. It emphasizes the need to shift toward modern, automated technologies to improve the effectiveness and efficiency of traffic management in increasingly complex urban environments.

Smart cities are a way to use technology to solve problems in cities. A smart city is an urban area built on ICT infrastructure that promotes sustainable development, efficient resource management, citizen participation, and innovation by integrating digital technologies to improve transportation, energy systems, public services, and overall quality of life (Waheeb & Wheib., 2024). The integration of smart technologies and intelligent transportation systems (ITS) improves the quality of life for residents, reduces congestion and pollution, and improves the efficiency and sustainability of urban areas (Babu et al., 2023). By using advanced monitoring systems that automatically detect traffic violations, like motorcycles entering bicycle lanes, cities can create safer streets for vulnerable road users and make better use of current infrastructure, helping move toward more sustainable urban mobility.

Cities sustainability increasingly depends on effective traffic management systems that can reduce congestion, lower emissions, and create safer environments for all road users. According to the United Nations' Sustainable Development Goals (SDGs), particularly Goal 11 (Sustainable Cities and Communities), By 2030, cities must provide access to safe, affordable, accessible and sustainable transport systems for all, improving

road safety, notably by expanding public transport, with special attention to the needs of those in vulnerable situations, women, children, persons with disabilities and older persons (United Nations, 2015). By using automated systems to monitor and enforce proper lane usage, cities can create safer and fairer for everyone.

Deep learning technologies, especially those focused on computer vision, have advanced significantly in recent years. According to Sohan et al. (2024), object detection is a vital task in computer vision, which is widely used in various fields such as robotics, medical imaging, surveillance systems, and autonomous vehicles. Object detection frameworks like YOLOv8 excels in identifying and classifying objects across various complex environments, showcasing significant improvements in accuracy, speed, and efficiency compared to other popular models like Faster RCNN, SSD, and RetinaNet. In a study conducted by Tahir et al. (2024), researchers developed an enhanced model called PVswin-YOLOv8s, which integrates Swin Transformer, CBAM, and Soft-NMS to improve detection of small, occluded, and overlapping objects. This hybrid model achieved a 4.8% higher detection accuracy than the baseline YOLOv8s on the VisDrone2019-val dataset, demonstrating YOLOv8s's adaptability and potential for high-performance detection tasks. Similarly, YOLOV8 has been used with EasyOCR for vehicle number plate detection where the model successfully detected and captured vehicle license plates from live video footage with an accuracy of 86% (Patil et al. 2023). These advancements in object detection not only enhance visual recognition in general computer vision applications but also lay the foundation for more specialized technologies just like in the part of Automatic License

Plate Recognition (ALPR).

Automatic License Plate Recognition (ALPR) plays a crucial role in modern traffic monitoring systems, helping enforce laws by addressing issues like vehicle theft, unauthorized access, and traffic violations (Indira et al., 2019). The integration of these technologies with current traffic monitoring systems could offer a flexible way to solve the issue of motorcycles entering pedestrian and bicycle lanes. Modern ALPR systems that use open-source OCR tools EasyOCR provide a cost-effective and accessible solution for license plate recognition tasks. In a recent study, (Salsabila & Sriani 2024) combined YOLOv8 for object detection, CNN for feature extraction, and EasyOCR for text recognition achieved a 74.66% average text reading accuracy. Likewise, (Vedhaviyassh et al., 2022) highlights that EasyOCR outperforms Tesseract OCR in license plate recognition, achieving over 95% accuracy compared to Tesseract's 90%. They also noted that EasyOCR deep learning approach enhances its efficiency in real-time predictions, making it suitable for various applications.

This study aims to develop and evaluate the use of deep learning techniques to automatically detect motorcycle intrusions on bicycle lanes, as well as recognize license plates to assist in enforcing traffic rules. The proposed model will leverage advanced computer vision techniques to accurately identify motorcycles that are improperly using these designated lanes. By extracting license plate information, the system will provide a comprehensive solution for monitoring compliance with lane usage regulations. This approach not only enhances safety for cyclists but also promotes adherence to traffic laws by enabling efficient identification and

follow-up on violators. Ultimately, the study seeks to create a robust framework that supports better urban mobility management and improves safety in shared roadway environments.

Objectives of the Study

The foremost objective of the study is utilizing deep learning to develop a model by optimizing YOLOv8 to detect motorcycle intrusion and recognize license plate entering the designated bicycle lanes. Further, the study aims to accomplish the following:

1. To collect and annotate images of motorcycles, bicycle lanes, Philippine motorcycle license plates, and other road vehicles on Roboflow, apply preprocessing techniques and prepare segmented bicycle lane data as region of interest(ROI).
2. To develop and train a detection model that identifies motorcycle intrusion using YOLOv8 and U-Net with ResNet34 for segmentation, and to apply EasyOCR to extract and recognize the alphanumeric characters found on Philippine motorcycle license plates.
3. To evaluate the performance of the optimized YOLOv8 and U-Net models in detecting motorcycle intrusions, recognizing license plates, and accurately segmenting bicycle lanes. Analyzing key performance metrics such as Precision, Recall, F1 Score, and Mean Average Precision for vehicle detection, alongside the Dice coefficient and Intersection over Union (IoU) for bicycle lane segmentation. Additionally, the study evaluates EasyOCR performance based on the Character Recognition Rate (CRR) and Plate Recognition Rate (PRR). Lastly, to validate model's performance through statistical method.

Significance of the Study

The study seeks to address the issue of motorcycle intrusion in pedestrian and bicycle lanes by utilizing YOLOV8 and EasyOCR for detection of motorcycle intrusion and license plate recognition. This research has the potential to enhance traffic enforcement and promote road safety.

This research aligns with the United Nations Sustainable Development Goals (SDGs), particularly Goal 11: Sustainable Cities and Communities, and Goal 3: Good Health and Well-being. UNESCO emphasizes the importance of improving urban infrastructure and reducing traffic-related injuries, which this study directly addresses.

Local Governments and Traffic Authorities. This study helps local governments and traffic authorities with real-time monitoring tools for detecting motorcycle intrusions in restricted lanes. By automating the identification of violators through license plate recognition, it streamlines the enforcement of traffic laws. This allows for quicker response times and reduces the reliance on manual monitoring. The collected data can be used for traffic analysis and policy development. It enhances operational efficiency and promotes safer, more organized roads.

Motorcyclists. In this study, motorcyclists are the focus of detection and regulation to ensure they do not intrude into restricted areas such as bicycle- designated lanes. By training a model to identify and track motorcycle intrusions, the study aims to support the enforcement of traffic laws and promote lawful riding behavior, thereby helping to prevent accidents and enhance overall road safety.

Bicyclists. For non-motorized two-wheeled vehicles, study prioritizes their safety by reducing the unauthorized presence of motorcycles in lanes exclusively designated for bicycles. With fewer violations, bicyclists can navigate these dedicated spaces with increased confidence and lower risk. This contributes to a more inclusive, cyclist-friendly urban environment and supports a shift toward sustainable, healthier modes of transportation.

Future Researchers. Future researchers can use this study as a foundation to enhance license plate recognition technologies and explore new innovations in traffic management. The findings and methodologies may inform the development of more accurate and efficient deep learning models. Researchers can also analyze the impact of such systems on traffic behavior and urban planning. This project opens avenues for interdisciplinary studies involving computer vision, smart cities, and public policy. It encourages continued progress in intelligent transportation systems.

Scope and Limitations of the Study

In developing a deep learning model, study focuses on detecting motorcycle intrusions on bicycle lanes and performs license plate recognition. YOLOV8 for object detection, U-Net for bicycle lane segmentation, and EasyOCR for license plate character extraction. Although the dataset includes other vehicles like cars and trucks since they are also potential violators, it is not the main focus of the model. These additional classes are included to help reduce false detections and improve the model's ability to distinguish motorcycles from other vehicles, but their violations are not included and evaluated. EasyOCR results was evaluated by calculating the Character Recognition Rate (CRR) and Plate Recognition Rate (PRR) using a subset of

the license plate dataset. This is due to the fact that some images contain similar visuals and identical license plates, as they were extracted from video footage. The study is limited to urban areas in the Philippines, where the dataset was collected and tested under clear, daytime conditions. The dataset used in this study was captured from a single camera angle, which limits the model's ability to recognize motorcycle intrusions from different viewpoints or perspectives. This geographic and environmental constraint may affect the generalizability of the model to other regions or scenarios, such as nighttime or poor weather. The model's accuracy and reliability may also be affected by external factors like lighting conditions, camera quality, and object occlusion, especially in real- world environments where ideal conditions are not guaranteed.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE STUDY

Review of Related Literature

This chapter provides key findings of related literature necessary to support and validate the development of a research study entitled “Utilization of Deep Learning: Optimized YOLOV8 For Detection of Motorcycle Intrusion and License Plate Recognition on Bicycle Lanes”.

Deep Learning for Computer Vision

Deep learning and computer vision integration yields superior performance in image classification and object detection tasks, significantly surpassing traditional methods. According to Liu et al. (2023) deep learning significantly enhances computer vision by enabling end-to-end image analysis, hierarchical feature extraction, and semantic understanding. They excel in tasks like image classification and object detection, surpassing traditional methods and driving advancements in intelligent vision systems.

Deep learning algorithms for computer vision include Convolutional Neural Networks (CNN), Restricted Boltzmann Machines (RBM), Stacked RBM, and Extreme Learning Machines. These techniques are widely used for applications such as image classification, object recognition, and detection (Voulodimos et al., 2018; Mahmoud & Ahmed, 2024). Over the years, deep learning architectures have significantly evolved. Starting from LeNet (LeCun, 1998) in the 1990s to more complex models like AlexNet (Krizhevsky et al., 2012), VGGNet, ResNet, and EfficientNet (He et al., 2016; Tan & Le, 2019; Dhillon & Verma 2020), these advancements have enhanced both accuracy and processing efficiency in vision-related tasks.

In object detection, there are two main types of deep learning architectures: two-stage detectors and one-stage detectors.Two-stage and single-stage detectors are the two basic categories into which object detectors can be roughly divided (Patil et al., 2024). These include two-stage models such as R-CNN, Fast R-CNN, and Faster R-CNN and single-stage models YOLO and SSD (Shetty, 2021). Models like R-CNN, Fast R-CNN, and Faster R-CNN have significantly influenced the field. These models use region proposals to detect and classify objects with high precision (Ren et al., 2015). However, they have limitations such as slower inference speeds, increased computational complexity, and the need for additional training steps. To address these limitations, more efficient algorithms like single-stage detectors have emerged (Varghese and Sambath 2023). Single stage detectors like YOLO (You Only Look Once) and SSD (Single Shot MultiBox Detector) solve this problem by detecting and classifying objects at the same time throughout the whole image. The YOLO models, created by Redmon and his team, split the image into a grid and directly predict bounding boxes and what types they are (Redmon et al., 2016).

Later versions of YOLO, like YOLOv2, YOLOv3, YOLOv4, and the newest one, YOLOv12, have become faster and more accurate, this makes them great for real-time applications, such as traffic monitoring and license plate detection. The study by Gorak and Reddy (2024) showed that YOLOv5 and YOLOv8 are effective in various real-world applications, such as automated toll collection and traffic monitoring. Similarly, Vijayakumar and Vairavasundaram (2024) demonstrated the ability of Yolo to perform real-time object detection with high accuracy by processing images in a single pass,

significantly improving efficiency compared to traditional methods like R-CNN and Viola-Jones. In line with this, Santos et al. (2021) noted YOLOv3 is effective due to its speed and ability to distinguish visually similar objects like motorcycles and bicycles, particularly with high-quality labeled datasets. Similarly, YOLOv8 is also used in systems detecting helmet violations and triple riding in real-time traffic videos, demonstrating robust accuracy under various real-world scenarios (Kumar et al., 2024).

Transfer learning has become an important method for using deep learning models in specific tasks when there is not much training data available. According to (Letho & Potter, 2023; Egon & Karl, 2023) transfer learning and pre-trained models have revolutionized the field of deep learning, enabling the development of highly accurate and efficient machine learning systems, even when faced with limited data and computational resources. Using a pre- trained model speeds up the training process as the model already knows important features related to the task, like shapes and patterns in images. In LPR, pre-trained convolutional neural networks (CNNs) can be fine-tuned on specific license plate datasets, enhancing accuracy and robustness (Maurya, 2023).

Traffic Monitoring Systems

Vehicle classification is an important part of today's traffic monitoring systems. It helps us analyze traffic patterns in detail and enforce rules more effectively. According to Shokravi et al. (2020), vehicle classification (VC) is an underlying approach in intelligent transportation systems and is widely used in various applications like the monitoring of traffic flow, automated parking systems, and security enforcement. Traditional vehicle classification

methods, as noted by Gholamhosseinian and Seitz (2021) rely on intrusive methods (like loop detectors and magnetic sensors) and non-intrusive sensors such as cameras and acoustic sensors. Each type has limitations, such as environmental sensitivity and lack of real-time data. These methods often require upgrades through advanced technologies like VANETs, which can provide more comprehensive data through vehicle-to-vehicle and vehicle-to- infrastructure communication to enhance accuracy and efficiency.

Detection and classification of vehicles are particularly challenging in video-based intelligent transportation systems. Traditional methods, as noted by Rashid (2010), depend on background subtraction from video frames to detect moving objects. However, these methods are computationally expensive and often perform poorly when vehicle and background pixel intensities are similar, leading to inaccurate detection and classification.

Computer vision has greatly improved how we classify vehicles. It has evolved from traditional methods to modern deep learning techniques. With the rise of deep learning, especially convolutional neural networks (CNNs), classifying vehicles has become much more precise. Vijayaraghavan (2019) discusses the effectiveness of Convolutional Neural Networks (CNNs) in vehicle classification, showing that CNNs can accurately detect and distinguish different vehicle types, even when they vary in size. Similarly, Trivedi et al. (2021) highlights the role of CNNs in vehicle classification by automatically identifying key features and achieving high accuracy. The results show that CNNs can achieve up to 99% accuracy in some categories, proving their effectiveness for real-time applications in intelligent transportation systems.

Additionally, Watkins et al. (2018) explored the use of ResNet-18, ResNet-34, and ResNet-50 to classify different types of vehicles on the Comprehensive Cars dataset. By integrating Spatially Weighted Pooling and a localization step, they improved the model's performance, achieving a top accuracy of 96.35%, proving that ResNet models are very effective for recognizing specific vehicles.

Along with this, Espinosa et al. (2018) compared several models, such as Espinet, CNN, and YOLO, for motorcycle detection, with Espinet achieving the highest precision at 93.7%. These studies commonly reported challenges in detecting small and obscured motorcycles, emphasizing the importance of selecting appropriate techniques to improve detection accuracy. Shoman et al. (2024) developed a helmet detection framework using YOLOv8, DCGANs, and Test Time Augmentation to address class imbalance and improve performance under varying weather and lighting conditions, with YOLOv8 yielding the best results. Meanwhile, Zhang et al. (2024) introduced a nighttime vehicle detection method utilizing enhanced HOG features, SVM, and a Kalman filter, which improved accuracy in low-light environments.

In the Philippines, a study by Cruz et al. (2022) used the YOLOv4 algorithm to successfully detect and count motorcycles and vehicles in real-time traffic videos from Katipunan Avenue Southbound, Metro Manila. The system, which included video annotations and class counters, proved effective for traffic monitoring and supported the development of motorcycle-inclusive road safety systems.

Despite improvements, Masula et al. (2024) emphasized of identifying motorcycles in various traffic situations remains challenging due to their

diverse designs and sizes, which complicate accurate detection. Regular detection methods often struggle with motorcycles, especially in crowded scenes, where their small size and varying appearances depending on camera angle and distance make recognition difficult. Their study showed that both the CNN model and the YOLOv2 network achieved high accuracy in motorcycle recognition; the YOLOv2 model had an accuracy of 99.67%, while the CNN model developed by the authors achieved an accuracy of 99.56%.

Intrusion Detection in Restricted Zones

Intrusion detection in restricted zones is a critical component of modern surveillance and autonomous driving systems. In urban road environments, illegal intrusion by pedestrians and cyclists into restricted lanes presents a serious safety issue. A novel solution is presented in the Object-Level Phase Space Reconstruction Network (PSRNet), which classifies motion time series to detect lane intrusions up to 150m ahead using a monocular camera. By transforming object movement into latent phase space trajectories, PSRNet employs a Lyapunov exponent-like classifier for accurate action recognition, achieving 98.0% accuracy on the THU-IntrudBehavior dataset—surpassing traditional recognition models by over 30% (Zhang et al., 2021).

For border and restricted area security, a YOLO-based system has been developed to detect human intruders in real-time. The model achieves approximately 96% accuracy across varying environmental conditions, effectively reducing the need for risky manual inspections (Chandra & Panda 2021). In urban transit, unauthorized vehicle entry into Bus Rapid Transit (BRT) lanes undermines efficiency. A system utilizing YOLOv8 and EasyOCR

identifies such vehicles and extracts license plates to verify access, reaching 80% accuracy using a hybrid dataset (Garcia et al., 2023).

Detection of vehicles in a pedestrian pathway is important for the safety of pedestrians (Virumandi et al., 2015). As autonomous vehicles become more common on the roads, their advancement draws on safety concerns for vulnerable road users, such as pedestrians and cyclists (Ahmed et al., 2019). In the context of the Philippines, a study in Marikina City revealed that vehicle intrusion into bike and pedestrian lanes is a growing concern. The findings showed that 58.8% of respondents encountered conflicts with vehicles in bike lanes, while 52.9% noted frequent obstructions from parked cars. This highlights the urgent need for vehicle intrusion detection systems to ensure the safety of cyclists and pedestrians (Camba et al., 2017).

Recent research has specifically addressed the detection of vehicles intruding into restricted lanes such as bus lanes, bicycle paths, and pedestrian zones. Pajuelo and Peredo (2024) developed a system to detect and classify different types of vehicles that enter bicycle lanes. They used deep learning techniques, specifically the YOLO (You Only Look Once) models, testing versions from YOLOv4 to the newest YOLOv8. They found that the YOLOv8m model worked best in their tests, achieving an F1 score of 88.00% and a mean Average Precision (mAP) of 94.80% at a 0.50 IoU threshold. This model was effective in identifying five types of vehicles: bicycles, motorcycles, electric motorcycles, motorcycle taxis, and electric scooters. The goal of this research is to make roads safer for cyclists and to improve traffic management in cities. Similarly, Hernández-Díaz et al. (2024) created a system that uses deep learning, specifically YOLOv8, to detect motorcycle

violations in pedestrian crosswalks, trained on a new dataset of over 6,000 annotated traffic images categorized into four classes. Their results showed an average accuracy of 84.6%, which shows that computer vision models can help improve traffic enforcement and make pedestrian areas safer.

Region of Interest (ROI) extraction is essential for monitoring specific areas within surveillance footage, especially for detecting intrusions in restricted zones like bicycle lanes . The research by Hoang etal. (2019) emphasizes that ROI extraction plays a vital role in the detection and recognition of road markings. By focusing on key areas within the image, ROI extraction improves the efficiency and accuracy of road marking detection , ensuring that only relevant information is processed, which is particularly beneficial in real-time applications.

From a recent study by Antwi et al. (2024) introduced the innovative approach using deep neural network models on image processing and computer vision to be able to detect and extract pedestrian and bicycle lane features from very a high-resolution aerial imagery that focuses in public roadways in Florida. When tested against ground truth data from Leon County, Florida, the models demonstrated accuracy rates of 73% for the pedestrian lanes and 89% for bicycle lanes, highlighting the effectiveness of precise Region of Interest (ROI) extraction in enhancing detection performance. Additionally, Lu (2015) proposed a simplified module for lane detection that utilizes Maximally Stable Extremal Regions (MSER) segmentation and the Progressive Probabilistic Hough Transform (PPHT) to improve detection accuracy. This approach achieves an average detection rate of 92.7%, significantly improving accuracy compared to traditional methods, with

detection rates exceeding 95% under complex urban conditions, demonstrating the value of effective ROI extraction in real-time lane detection systems.

Several studies utilize U-Net, a model developed by Ronneberger et al. (2015) for medical image segmentation, because of its special design with encoder- decoder architecture and skip connections that allows for accurate pixel-by- pixel segmentation. This feature makes U-Net very useful for intrusion detection in restricted zones, as it can clearly detect lane markings and intrusions with detailed accuracy, making it perfect for real-time monitoring in urban areas. Jayakumar et al. (2024) presents a U-Net-based lane detection model optimized through hyperparameter tuning and five-fold cross-validation, achieving a high training accuracy of 99.93%, validation accuracy of 99.81%, and test accuracy of 99.78%, with an IoU score of 0.8861 and F1 score of 0.9396, demonstrating excellent performance and generalization to new data. The study by Tangestanizadeh et al. (2024) compared two deep learning models for lane detection, with the attention-enhanced U-Net model achieving a high accuracy of 98.98%, significantly outperforming the FPN model's 87.59%. These results highlight the effectiveness of incorporating attention mechanisms into U-Net for more accurate and reliable lane detection, especially in autonomous driving systems.

Study of Hotkar et al. (2023) highlights that U-Net is effective for accurately detecting lane markings using the CULane dataset, while YOLO offers high accuracy and low complexity in object detection, making both suitable for real- time autonomous driving applications. It also emphasizes the

models' potential in improving transport safety, while noting challenges such as detection accuracy and latency that need to be addressed in future work. These studies show U-Net's versatility and effectiveness in lane detection tasks, which can be adapted for bicycle lane detection.

Automatic License Plate Recognition

Automatic License Plate Recognition (ALPR) or Automatic Number Plate Recognition (ANPR) is the technology responsible for locating and extracting the license plates of a vehicle in an image or a video sequence using the optical character recognition (OCR) (Ahmad et al., 2015; Ravichandran, 2025). More so, the extracted information can be used with or without a database in many applications, such as electronic payment systems (toll payment, parking fee payment), and freeway and arterial monitoring systems for traffic surveillance (Du et al., 2013).

Traditionally, the Automatic License Plate Recognition (ALPR) systems primarily rely on image processing techniques to identify and extract license plate information from vehicles. Sanogo et al. (2019) discussed the Traditional ALPR systems follow a step-by-step process: detecting the license plate, pre-processing the image, segmenting characters, classifying them, and applying post-processing. However, these systems often struggle in challenging situations easily affected by conditions of light, shadow, background complexity or other factors, resulting in the failure to meet the application of real scenes (Weihong & Jiaoyang, 2020). According to Duan (2005) Automatic license plate recognition (ALPR) has been studied since the 1990s, with early methods focusing on boundary line features. These approaches used techniques like gradient filters to enhance edges, followed by

binarization and line detection using algorithms such as the Hough Transform. In the study of Ahmed et al. (2003) Hough Transform was used to extract licence plates but the disadvantage is that this method requires a huge memory and is computationally expensive.

In modern ALPR, it uses computer vision and machine learning techniques to identify and extract license plate details from images. In the study of Ahmad et al. (2024) Automatic Number Plate Detection uses machine learning to convert images into text by segmenting the photo, locating and extracting the vehicle and its number plate, then recognizing characters using OCR. This technology is applied in areas like military zones, apartment buildings, and traffic surveillance. Moreover, these systems plays a significant role in border surveillance, ensuring the safeguards, and handling vehicle-related crime. Hence, the most effective approach for implementing ALPR systems utilizes deep learning via convolutional neural network (CNN). A CNN works on an input image by assigning significance to various features of the image and differentiating them from each other (Kaur, 2022). In the study of Laroca (2018) Convolutional Neural Networks (CNNs) are trained and fine-tuned for each ALPR stage to handle varying conditions like different cameras, lighting, and backgrounds. For character segmentation and recognition, a two-stage method is used with data augmentation techniques, including inverted license plates and flipped characters. According to Jørgensen (2017) high prediction accuracy and speed of ALPR is vital. Recent improvements in deep learning have increased its ability to solve complex visual recognition tasks. Using deep learning to improve the accuracy and speed of solving the ALPR task is for this reason promising.

Several Architectural innovations have been created just for ALPR applications. Multi-stage networks, which separate detection and recognition tasks, have been especially effective. Dhedhi et al. (2018) presents an ALPR system that combines YOLO and CNN. It uses three YOLO stages for detecting and segmenting the license plate, and one CNN stage for recognizing the characters. The system was trained using 604 real car images and 42,237 computer-generated characters. On 50 test images, it achieved an overall accuracy of 82%, with each stage reaching over 90% accuracy in validation.

For recognizing motorcycle license plates, new designs have been created that improve the ability to identify smaller targets. Nong et al. (2020) developed an intelligent system using FPGA for motorcycle plate detection, employing the Sobel Edge Detection Algorithm. It achieved recognition rates of 83.3% for low-resolution images and 50% for high-resolution images, effective for both still and moving conditions.

Despite significant advances, ALPR systems still face several challenges in real-world scenarios. According to Sultan et al. (2023) License plate recognition (LPR) plays a key role in modern intelligent systems, but it remains a difficult task due to diverse plate shapes, non-standard templates, irregular outlines, varying angles, and occlusions, all of which hinder accurate detection and recognition. Shashirangana et al. (2020) also highlights similar difficulties, emphasizing the complexities in handling these variations that significantly impact the effectiveness of LPR systems.

For motorcycle license plates, specific challenges include the occlusion, poor illumination, low-quality surveillance footage, and the fluctuating weather

conditions, which complicate accurate detection and recognition (Devi et al., 2022). Further, Padalia et al. (2023) identify additional problems, such as the complexity of detecting motorcycles in crowded traffic, handling overlapping motorcycles, ensuring real-time detection of both motorcycles and license plates, and handling different image quality from CCTV footage. These challenges show the complexity of developing efficient and accurate systems for motorcycle license plate detection and recognition.

Literature Matrix

Below is the list of reviewed related literature relevant to the development of the study regarding the detection of motorcycle intrusion and license plate recognition on bicycle lanes.

Table 1. Matrix of Reviewed Related Literature

AUTHOR/S	TITLE	YEAR	ALGORITHM / TECHNIQUE	FINDINGS
Gorak and Reddy	Helmet and Number Plate Detection Using YOLOV5 and YOLOV8	2024	YOLOv5, YOLOv8	Both are effective in toll collection and traffic monitoring.
Pajuelo and Peredo	Comparison and evaluation of YOLO models for vehicle detection on bicycle paths	2024	YOLOv4–YOLOv8	Augmented dataset using GANs; YOLOv8 performed best under varied conditions.
Hernández-Díaz et al.	A computer vision system for detecting motorcycle violations in pedestrian zones	2024	YOLOv8	Achieved 84.6% average accuracy with 6,000+ annotated images.
Ahmad et al.	Deep Learning for Computer Vision: Innovations in Image Recognition and Processing Techniques	2024	OCR, ML	Converts images to text with applications in surveillance and border security.

Antwi et al.	Automated Detection of Pedestrian and Bicycle Lanes from High- Resolution Aerial Images by Integrating Image Processing and Artificial Intelligence (AI) Techniques	2024	Deep Neural Networks	Achieved 89% accuracy for bike lanes and 73% for pedestrian lanes using high-resolution imagery. Supports effective ROI extraction.
Mahmoud & Ahmed	Deep Learning for Computer Vision: Innovations in Image Recognition and Processing Techniques	2024	CNN, RBM, Stacked RBM, ELM	These techniques are effective in image classification and object recognition.
Chandra &	A human intruder detection system for restricted sensitive areas	2021	YOLO-based	96% accuracy in detecting human intruders in varying environmental conditions.
Trivedi et al.	Motorcycle detection and classification in urban Scenarios using a model based on Faster R- CNN	2021	EasyOCR , Tesseract	EasyOCR achieved 95% vs Tesseract's 90%. Suitable for Philippine motorcycle plates
Shokravi et al.	A review on vehicle classification and potential use of smart vehicle-assisted techniques	2020	Vehicle Classification	Vehicle classification enables traffic pattern analysis and enforcement.
Hoang et al.	Enhanced detection and recognition of road markings based on adaptive region of interest and deep learning	2019	ROI extraction	ROI extraction improves road marking detection by focusing on key image areas.
Laroca	A robust real- time automatic license plate recognition based on the YOLO detector	2018	CNNs, data augmentation	CNNs are effective across various conditions and improve ALPR performance.

Espinosa et al.	Motorcycle detection and classification in urban Scenarios using a model based on Faster R- CNN	2018	Espinet, CNN, YOLO	The Espinet model achieved the highest precision at 93.7%.
Dhedhi et al.	Automatic License Plate Recognition Using Deep Learning	2018	YOLO, CNN	Combines YOLO for detection and CNN for recognition, achieving high accuracy
Voulodimos et al.	Deep learning for computer vision: A brief review	2018	CNN, RBM, Stacked RBM, ELM	These techniques are effective in image classification and object recognition.
Redmon et al.	YOLO: Real- Time Object Detection	2016	YOLOv1	YOLO performs simultaneous object detection and classification, suitable for real- time applications.

Synthesis and Relevance of the Study

The collected literature emphasizes the rapid advancement of deep learning algorithms, particularly YOLO (You Only Look Once) models and Convolutional Neural Networks (CNNs), in real-time object detection and automatic license plate recognition (ALPR). Across various studies from 2016 to 2024, YOLO variants (YOLOv1 to YOLOv8) have demonstrated high accuracy in detecting vehicles, helmets, intrusions, and license plates under diverse conditions. In particular, YOLOv8 consistently outperforms earlier models in precision and efficiency. Other approaches like OCR, stacked RBMs, and deep neural networks also contribute significantly to surveillance, classification, and image recognition tasks.

The literature also explores the integration of AI with aerial imaging, region of interest (ROI) extraction, and enhanced image processing to

improve detection in complex urban and road environments. Techniques like data augmentation further strengthen the robustness of models in real-time applications.

In urban congestion and lane violations become more pressing, employing advanced deep learning techniques like YOLOv8 offers a practical, scalable, and accurate solution for monitoring and enforcing traffic laws. The high detection rates, real-time capability, and adaptability of these algorithms align well with the goal of identifying motorcycle intrusions in bicycle and pedestrian zones. Moreover, the studies reinforce the viability of integrating ALPR systems with traffic surveillance, contributing to safer and smarter urban mobility systems.

Conceptual Framework

This is the visual representation for the development of the study starting from the collection of raw images to the results of the study.

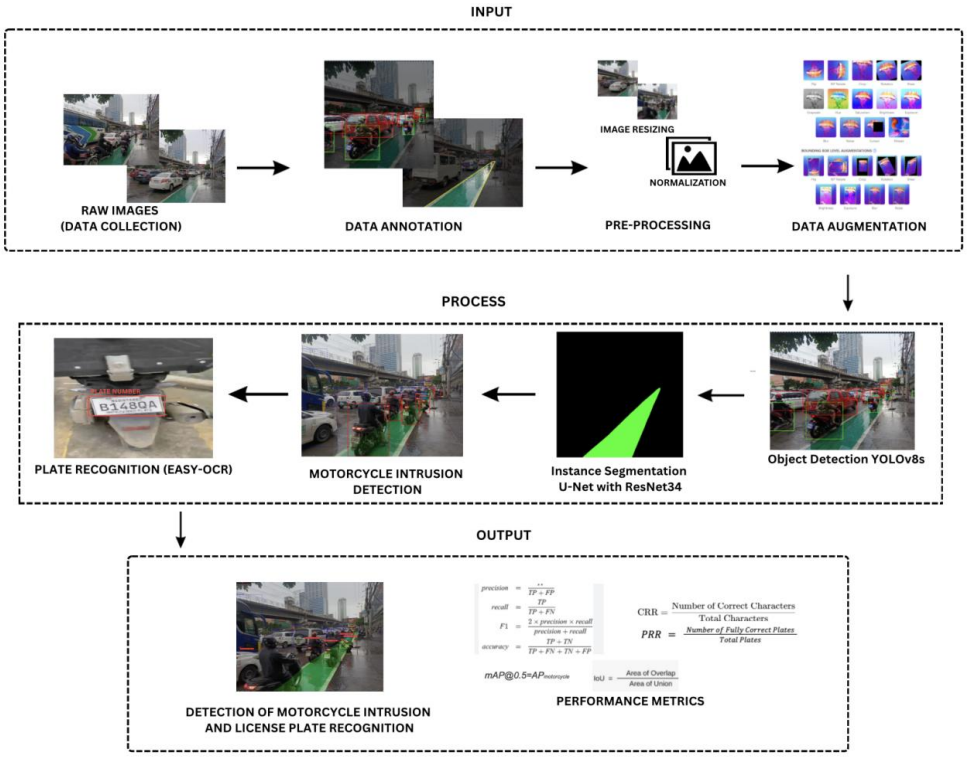


Figure 1. Conceptual Framework

CHAPTER 3

METHODOLOGY

Study Area

The study focuses on urban areas with clearly marked bicycle lanes, where motorcycle intrusions are commonly observed. Study area will be conducted in Cubao, Quezon City, Metro Manila. In this area, researchers captured sample images and videos of vehicles on the road adjacent to a bicycle lane. Data images were then used to create our datasets for training the model. This setting provided a rich variety of scenarios essential for addressing motorcycle intrusions in bicycle lanes. These locations offer a realistic environment for evaluating the model's performance under daytime conditions with fair visibility.

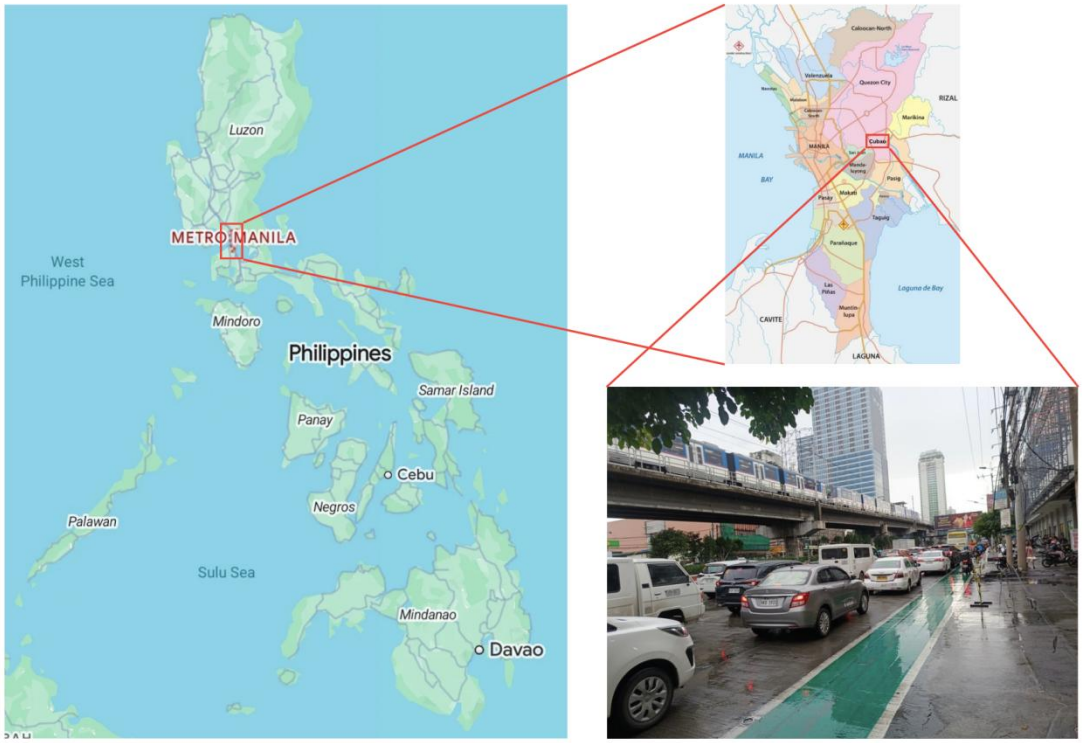


Figure 2. Location of the Study Area Used in the Model Development

Materials

The study used the following resources and materials, including software, hardware, and tools, which were essential to completing the model.

Software

The research used several software tools to help detect motorcycles in bicycle lanes and read their license plates.

Google Colaboratory (Colab)

Google Colaboratory (or Colab) serves as the primary development environment for this project. This platform provides access to high-performance computing resources, specifically Graphics Processing Units (GPUs), which significantly accelerate the training of detection models. Through Colab, the computer vision model was trained to recognize motorcycles, process license plate images, and evaluate performance using video recordings of bicycle lanes.

Visual Studio Code

Visual Studio Code functioned as the secondary development tool. It was utilized to develop programs for processing video footage, integrating motorcycle detection with license plate recognition components, and calculating model accuracy. This software facilitated efficient code organization and version control throughout the research process.

Roboflow

Roboflow was employed to create and manage the image datasets required for the project. The researchers used this tool to label a total of 1,832 images. Additionally, the platform enhanced the dataset by automatically generating synthetic training images through augmentation techniques such as

flipping and rotation. These precisely labeled images were essential for training the computer vision model to accurately detect motorcycles and license plates.

JAMOVI

JAMOVI was utilized for the statistical analysis of the gathered data. This open-source software provided a user-friendly interface for conducting various statistical tests and visualizing data distributions, aiding in the interpretation of the model's performance results.

JASP

JASP complemented the analytical phase by allowing for advanced statistical modeling and Bayesian analysis. Its intuitive layout enabled the exploration of complex analyses without the need for extensive programming knowledge, thereby enhancing the overall understanding of the data.

OpenCV

OpenCV (Open Source Computer Vision Library) is a vital library for image processing and computer vision tasks. In this research, OpenCV was used to manipulate and process images from video footage, enabling tasks such as filtering, edge detection, and image transformations. Its extensive functionalities allowed for the effective preparation of images for the YOLOv8 model.

Ultralytics YOLOv8

The Ultralytics YOLOv8 framework served as the backbone of the motorcycle detection model. As a state-of-the-art object detection algorithm known for speed and accuracy, YOLOv8 was optimized to enhance the detection of motorcycles and license plates within bicycle lanes, leveraging its capabilities for real-time applications.

U-Net

A deep learning architecture primarily used for image segmentation, was implemented to isolate bicycle lanes from the background in video sequences. By accurately defining lane boundaries, the model's ability to distinguish between motorcycles and the surrounding environment was significantly improved.

EasyOCR

EasyOCR, a Python library for optical character recognition, was utilized to recognize and extract text from license plates detected in the video footage. Its capacity to handle various fonts and formats made it an ideal choice for the accurate extraction of license plate information.

NumPy

A fundamental library for numerical computing in Python, NumPy was essential for handling arrays and performing mathematical operations on large datasets. In this research, NumPy enabled efficient data manipulation and processing, supporting the study's overall computational workflow.

Pandas

Pandas is a powerful data manipulation and analysis library used for handling structured data. The researchers utilized Pandas to manage and analyze collected datasets, facilitating essential tasks such as data cleaning, transformation, and summarization.

Matplotlib

A plotting library for creating static, interactive, and animated visualizations, was used to visualize the results of the analyses. It assisted in presenting performance metrics and model predictions, effectively illustrating the study's findings.

Hardware

The development and testing of the motorcycle intrusion detection system were conducted using the following hardware specifications:

Table 2. Desktop Specification

Minimum Desktop Specification	
Device name	LAPTOP-516J412T
Processor	12TH Gen Intel(R) Core (TM) i5-1240P 1.70 GHz
Installed RAM	8.00 GB
System Type	64-bit operating system, x64-based processor
Operating System	Windows 11 Home Single Language, Version 24H2 (Build 26100.3476)

Research Design

This study uses a quantitative experimental research design, focusing on computer vision and deep learning. The research aims to develop and test a model that can detect motorcycle intrusion and recognize license plate. The approach emphasizes careful data collection, thorough experimental procedures, and complete performance testing to ensure that the results are both scientifically valid and practically useful.

The research framework includes a clear three-stage process for analyzing images from phone cameras. In the first stage, the goal is to accurately identify bicycle lanes in traffic scenes. The U-Net architecture was used to segment and outline the boundaries of these lanes, ensuring they are correctly located. In the second stage, object detection techniques was applied to find and classify motorcycles within the identified bicycle lane areas. This step is important for ensuring that only the relevant vehicles are analyzed, which improves the model's overall efficiency. The final stage was optical

character recognition (OCR) technology to accurately read license plate text from the detected motorcycles. This part is crucial for gathering useful data from the identified vehicles.

The model utilizes a structured deep learning pipeline that incorporates validation stages to monitor performance. The process begins with image preprocessing to standardize resolution, color, and quality, thereby ensuring data consistency. This research design develops a robust system by orchestrating optimized YOLOv8 for motorcycle detection, U-Net with a ResNet backbone for lane segmentation, and integrated EasyOCR for license plate recognition (LPR)

Model Development

The model development was obtained by a systematic approach with three main components working together to create a motorcycle intrusion detection model. Each component is specially designed for a specific task and works in sequence to achieve the final goal. The bicycle lane segmentation part used U-Net architecture with ResNet34 encoder backbone, chosen for its proven excellent performance in segmentation tasks that need precise boundary detection. The U-Net architecture provides an effective encoder-decoder structure with skip connections that preserve fine details essential for accurate bicycle lane boundary identification. The ResNet34 encoder, pre-trained on ImageNet dataset, offers strong feature extraction abilities while maintaining computational efficiency suitable for real-time applications.

The model architecture processed the input images of 512×512 pixel resolution through the ResNet34 encoder to extract hierarchical feature representations at multiple scales. The decoder part reconstruct spatial

resolution through progressive upsampling operations while using skip connections to incorporate low-level features from corresponding encoder stages. The final output produced single-channel binary masks that precisely identify bicycle lane regions within the input images.

The motorcycle detection part used YOLOv8s (You Only Look Once version 8 Small) framework, specifically optimized for real-time object detection applications with balanced accuracy and computational efficiency. YOLOv8s uses an anchor-free detection mechanism that simplifies the architecture while maintaining competitive performance compared to anchor-based alternatives. The model process input images at 640×640 pixel resolution through a CSPDarknet backbone with Cross Stage Partial connections that enhance feature flow and gradient propagation.

The license plate recognition part used EasyOCR framework, which combines CRAFT (Character Region Awareness for Text detection) text detection with CRNN (Convolutional Recurrent Neural Network) text recognition for complete optical character recognition abilities. CRAFT provides robust text detection by generating character-level and word-level region scores, enabling accurate finding of license plate text regions even under challenging conditions such as varying orientations, lighting, and background complexity. The CRNN component performs character recognition through convolutional feature extraction followed by recurrent neural network processing that captures sequential dependencies between characters extraction followed by recurrent neural network processing that captures sequential dependencies between characters.

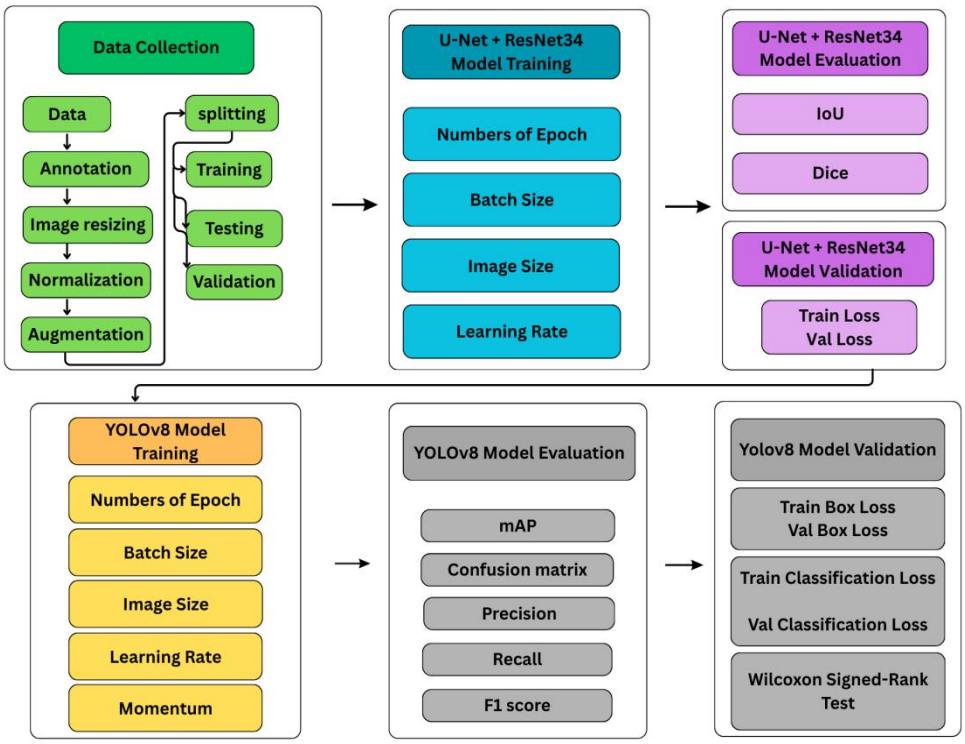


Figure 3.1 Model Development for Detection of Motorcycle Intrusion

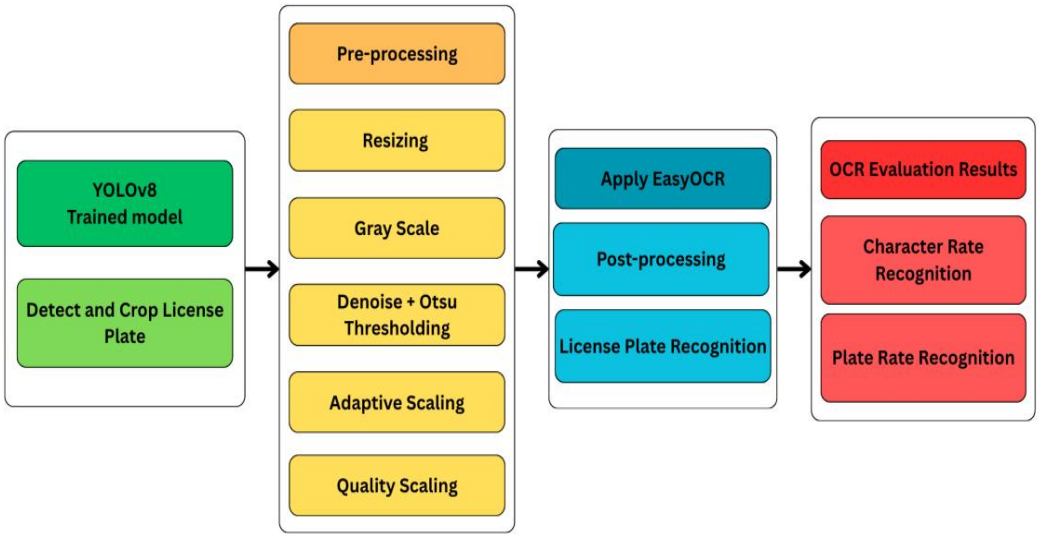


Figure 3.2 Model Development for License Plate Recognition

Data Acquisition

Datasets were collected and managed using Roboflow, a comprehensive computer vision platform that facilitates data collection, annotation, and pre-processing.

Table 3. Dataset Table

Types of Data Collected	Annotated Data
Motorcycle	10,358
other vehicles (e.g., cars & trucks)	12,655
License Plate	10, 325
Bicycle Lane	1,832
Total Instances	35,170

The research team collected a diverse dataset of 35, 170 annotated data from 1,691 raw images, which were augmented to 3,215 images for object detection and 1,832 images for image segmentation, further augmented to 2,841 images. Specifically, dataset includes 10,358 annotated motorcycles, 1,832 annotated bicycle lanes using polygon tools, 10,325 annotated license plates, and 12,655 annotated instances of other vehicles to help reduce false positives by teaching the model to differentiate between motorcycles and other vehicles. Data splitting use a ratio of 70-20-10 for training, validation, and testing. This comprehensive dataset is crucial for developing a model that can accurately detect and segment relevant objects.

All images were collected during daytime and annotations were performed using Roboflow built-in tools, applying bounding boxes for vehicles and license plates and polygon tools for bicycle lanes. To improve the robustness of the models, data augmentation techniques were applied through Roboflow, including horizontal flips, slight rotations, and brightness adjustments. These augmentations increased the variety of the training data and helped the models adapt to changes in angle, lighting, and perspective.

The data collection process prioritized variety and relevance while also

focusing on daytime conditions with good visibility as specified in the research scope. Roboflow's platform features facilitated version control of the dataset allowing for systematic tracking of changes and improvements throughout the research process. Platform's preprocessing capabilities were leveraged to prepare the images for model training, including resizing, augmentation, and normalization operations.

Model Selection

This research combines three specialized models to automatically detect motorcycle intrusions in bicycle lanes. Each model was carefully selected based on its strengths and how well it fits into the system's overall goals.

YOLOv8s is a real-time object detection model that identifies motorcycles in video frames. It was selected for its balance of speed and accuracy, making it ideal for live monitoring scenarios. YOLOv8 works by dividing each image into smaller regions and checking for objects across the entire frame. It is particularly effective at detecting small and fast-moving objects like motorcycles, even when they appear from different angles or under varying lighting conditions. Once a motorcycle is detected, the model draws a bounding box around it and assigns a confidence score to indicate how certain it is about the detection.

U-Net with Resnet34 is a deep learning model designed for image segmentation. Unlike object detection models that draw simple boxes, U-Net with Resnet34 provides pixel-level accuracy by generating detailed masks that outline specific areas in an image. In this system, U-Net with Resnet34 is used to detect and segment bicycle lanes. It is well- suited for this task

because it performs reliably even with a relatively small training dataset. U-Net with Resnet34 can accurately identify lane boundaries even if parts of the lane markings are faded, blocked, or poorly lit. The resulting segmentation mask is then converted into a binary mask to define the Region of Interest (ROI), allowing the system to focus only on lane areas where intrusions might occur.

EasyOCR is a lightweight and flexible optical character recognition (OCR) tool used to read license plate numbers from detected motorcycles. Once a motorcycle is identified inside a bicycle lane, the license plate region is cropped and preprocessed by converting it to grayscale, enhancing contrast, and applying noise reduction techniques to make the text more visible. EasyOCR then analyzes the cleaned image to extract readable characters. We chose EasyOCR because it supports multilingual text, requires minimal setup, and is compatible with real-world license plate formats, such as those used in the Philippines.

The model processes video footage frame by frame. First, U-Net with Resnet34 segments the bicycle lanes and creates a mask that shows where lanes are located. YOLOV8 then detects all motorcycles present in the frame. The system checks whether any motorcycle overlaps with the lane area defined by U-Net with Resnet34. If a motorcycle is detected inside the lane, it is flagged as an intrusion. The license plate of that motorcycle is then isolated, and EasyOCR is used to read the characters on it. All relevant information such as the time, plate number, and image frame is recorded for further review or enforcement.

By combining segmentation, object detection, and EasyOCR, model

offers a complete and automated solution for monitoring traffic violations on bicycle lanes. Each model plays a specific role in identifying where violations occur, detecting the responsible vehicle, and extracting key information like license plate numbers. This multi-model setup reduces false detections, improves accuracy, and eliminates the need for constant human supervision supporting safer roads for cyclists and more effective traffic rule enforcement.

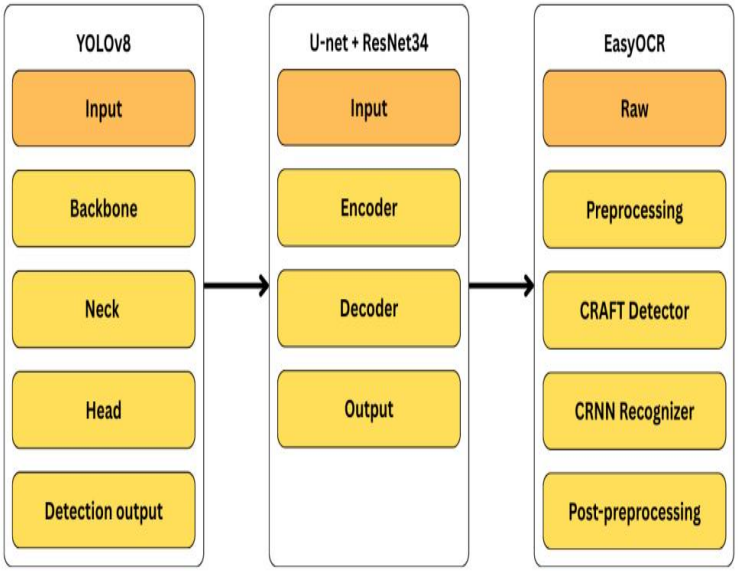


Figure 4. Model Architecture

Model Evaluation

This section presents the evaluation of the model using a set of performance metrics that provide insights into how well the system performs across its core tasks. These metrics help assess the model's accuracy, efficiency, and reliability in detecting motorcycle intrusions on bicycle lanes and recognizing license plates.

Vehicle and License Plate Detection (YOLOv8s)

The detection of unauthorized motorcycles on bicycle lanes is evaluated of the following metrics:

Precision. Measures the proportion of true positive detections among all posi-

tive predictions. It is used to assess how accurately the system identifies actual motorcycle intrusions without falsely flagging non-intrusions, such as bicycles or pedestrians.

Recall. Evaluates the model's ability to detect all actual motorcycle intrusions. It reflects how effectively the system captures every instance of intrusion, ensuring that violations are not missed.

F1-score. The harmonic mean of precision and recall, providing a balanced view of the model's detection performance. It is especially useful when both false positives and false negatives are critical, as in this study.

Mean Average Precision at IoU ≥ 0.5 (mAP@0.5). Assesses the model's ability to correctly localize and classify motorcycles based on bounding box overlap with ground truth. It is essential for verifying that the detected intrusions are not only identified but also accurately positioned within the frame.

Mean Average Precision at IoU ≥ 0.50-95. It checks model performance from lenient (50%) overlap at up to very strict (95%) overlap. A higher score means the model is both accurate in detecting objects and precise in bounding box placement.

Bicycle Lane Segmentation (U-Net with RestNet34)

IoU (Intersection over Union). This measures how much the predicted bicycle lane area overlaps with the ground truth lane mask. A higher IoU indicates that the model accurately segmented the bicycle lane boundaries, ensuring the Region of Interest (ROI) is correctly identified for detecting motorcycle intrusions.

Dice Coefficient (or F1 Score for segmentation). Similar to IoU but gives more weight to correctly predicted pixels. A high Dice score signifies that the predicted bicycle lane closely matches the actual lane area, which is essential for reliable detection and minimizing false segmentation of the ROI.

License Plate Recognition (EasyOCR)

Once a motorcycle intrusion is detected, the system performs license plate recognition to identify the vehicle involved. The performance of this component is measured using the following metrics:

Character Recognition Rate (CRR). CRR measures the percentage of individual characters correctly recognized from license plates. This metric helps assess how accurately the system reads each character, which is critical for partial plate matching and data entry reliability.

Plate Recognition Rate (PRR). PRR evaluates the percentage of complete license plates that are fully and correctly recognized. It provides a clear measure of the system's overall success in identifying entire license numbers without errors, which is essential for enforcement or tracking purposes.

Model Validation

Model validation was carried out to evaluate the generalization performance of the YOLOV8 model for vehicle detection and bicycle lane segmentation. A separate validation dataset, distinct from the training data, was utilized to assess the model's capability in detecting motorcycles and recognizing license plates under conditions it had not previously encountered. This step ensured that the model's performance was not limited to the training

data and could be reliably applied in practical settings, such as real-time monitoring of motorcycle compliance with traffic regulations.

The validation process involved tracking specific performance metrics that were automatically recorded by the training framework, which was based on a deep learning object detection architecture. These metrics were collected from the output logs of the training pipeline and are commonly used to assess object detection and classification models. Each of these validation metrics plays a significant role in interpreting how well the model performs in identifying, localizing, and classifying motorcycles.

Validation Box Loss. This metric quantifies the error in predicting the position and dimensions of the bounding boxes around detected motorcycles. It directly reflects the model's localization accuracy. Lower box loss values indicate that the model is learning to better fit the predicted bounding boxes to the actual positions of motorcycles in the image. This metric is crucial for precise localization, which is essential for effective monitoring of motorcycle intrusions.

Validation Classification Loss. This loss refers to the model's accuracy in correctly classifying detected objects as motorcycles or other vehicles. It helps determine whether the model can differentiate between motorcycles and non-target objects. In the context of this study, where the primary task is to detect motorcycle intrusions, this metric is highly relevant as it directly measures the model's identification capabilities.

Validation Distribution Focal Loss (DFL). Measures how confidently the model predicts the bounding box coordinates. It improves the quality of the predictions by focusing more on difficult-to-detect samples. This metric is

derived from recent advancements in object detection algorithms and is especially applicable to tasks requiring precise localization, such as motorcycle detection. Adding DFL enhances the reliability of bounding boxes, supporting future expansion of the system into tracking or behavior analysis.

Training Loss and Validation Loss. Measure how well the YOLOV8s model learns the features of the bicycle lane during training. Consistently decreasing loss values indicate effective learning of lane patterns and markings, while a large gap between training and validation loss may suggest overfitting, which could reduce accuracy in identifying the bicycle lane in real-world scenarios.

Learning Rate Progression. The learning rate was monitored during training and validation to ensure stable convergence of the model. This metric helps avoid issues like overfitting or vanishing gradients and ensures that the model optimizes effectively. Tracking the learning rate is standard practice in model training and was vital in fine-tuning the model's performance in this study.

Wilcoxon Signed-Ranked Test. if there is a significant difference between the baseline and optimized models, particularly when normality assumptions are not met. It evaluates whether model optimization has resulted in statistically significant improvements in performance metrics like F1 score, precision, and recall.

Ethical Consideration

It is necessary to mention that this section of the thesis addresses the essential ethical guidelines necessary to maintain research integrity and protect participants' rights in developing the motorcycle intrusion detection

model. The data collected for this study consists of video footage recorded on roads which is basically a data gathered in public. However, we remain committed to respecting privacy rights by ensuring that this data is used exclusively for the purposes of this research. Transparency about the use of this data is paramount, and we will only utilize the gathered footage in a manner that aligns with ethical standards, focusing solely on enhancing road safety through our thesis.

Furthermore, the evaluation of the model's performance will utilize appropriate metrics that offer a comprehensive understanding of its capabilities and limitations, ensuring that findings are not only precise but also contextually relevant. Accountability is paramount throughout the research, with clear documentation of methodologies and decisions made during data collection, model development, and evaluation. Finally, the study will report findings honestly and accurately, including any limitations or uncertainties in the methodology or results, thereby contributing positively to the field and upholding scientific integrity. These principles collectively ensure that the research is conducted responsibly and ethically, enhancing road safety while respecting individual rights.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the results and discussion of the overall research objectives, accompanied by visual representation of the findings. The study aims to optimized YOLOv8 for detection of motorcycle intrusion and license plate recognition on bicycle lanes.

Gathered Data

Data were gathered in Cubao, Quezon City, Manila, by recording motorcycle intrusions into bicycle lanes from an adjacent vehicle. Recorded video frames were uploaded to Roboflow for detailed annotation. This effort resulted in over 35, 170 annotated data points from 1,832 raw images, including motorcycles, license plates, and bicycle lanes. The data was significantly augmented and split into a 70% training, 20% validation, and 10% testing ratio, ensuring a robust and accurate model development.



(a) (b) (c)

Figure 5. Sample Data of (a) Raw Data, (b) Annotated Bicycle Lane, and (c) Annotated Vehicles and License Plate

Figure 5 shows the (a) Raw Data that displayed a raw image from dataset capturing vehicles on the road alongside a bicycle lane, illustrating the typical urban conditions where motorcycle intrusions may occur. Meanwhile, the (b) Annotated Bicycle Lane, and (c) Annotated Vehicles and License

Plates illustrate annotation process involved identifying and labeling various classes within the dataset that includes motorcycles, other vehicles, license plates, and bicycle lanes. For vehicles and license plates, bounding box was used to facilitate object detection while bicycle lanes were annotated using polygon tool for instance segmentation by accurately defining lane boundaries. After the annotation process, the images were pre-processed and exported, preparing the dataset for model development by optimizing YOLOV8s and employing U-Net with ResNet34 utilizing Google Colab. This comprehensive dataset is essential for developing an accurate and reliable model tailored to detect motorcycle intrusions in bicycle lanes.

Model Development Results

In the development phase of the study, researchers implemented deep learning models to perform detection and segmentation tasks. For detection, the model was designed to identify and locate specific objects within images, while for segmentation, it was designed to classify each pixel to differentiate regions such as lanes or vehicles. A well-prepared dataset with diverse samples was utilized to ensure the models could effectively learn visual patterns and spatial relationships. Additionally, EasyOCR was integrated during preprocessing to extract and recognize text information, specifically the characters on motorcycle license plates, without requiring additional model training. This phase served as the foundation for optimizing model performance, enhancing both accuracy and efficiency in identifying and analyzing relevant elements in the images.

Object Detection

Table 4 outlines the optimized hyperparameters used for training the YOLOv8 model, which are crucial for enhancing the learning process. The model is configured to train for 50 epochs with an input size increased from 640 pixels to 960 pixels for improved feature extraction. The AdamW optimizer, chosen for its adaptive learning capabilities, replaced the baseline SGD type. Weight decay was changed to 0.0001 to further mitigate overfitting, while MixUp and Copy-Paste augmentations are introduced to enrich the training dataset. The inclusion of label smoothing refine class predictions. By optimizing these hyperparameters, we significantly enhanced the YOLOv8 model's performance in detecting motorcycle intrusions and recognizing license plates, balancing memory efficiency and learning effectiveness to achieve higher detection accuracy and reliability.

Table 4. Optimized Hyperparameters of YOLOv8

Parameter	Baseline	Optimized
Model	YOLOv8s	YOLOv8s
Image Size	640px	960px
Optimizer	SGD (default)	AdamW
Learning Rate	Default (0.01)	0.001
Weight Decay	None	0.0001
Augmentation	Basic	MixUp + Copy-Paste
Regularization	None	Label Smoothing
LR Scheduler	Default	Cosine
Epochs	50	50

Figure 6. Showcase the model's ability to detect various vehicles and license plates in urban environments. Each image featured bounding boxes around detected objects, labeled with classes such as "vehicles," "motorcycles," and "license plates," alongside confidence scores ranging from 0.3 to 0.8. The model demonstrated strong performance in identifying multiple vehicles, though some detections show lower confidence scores, likely due to factors like occlusion, low image quality, or challenging lighting conditions. To sum it up, these results highlight the efficiency of the YOLOv8 model in real-world vehicle detection scenarios, emphasizing its potential for applications in monitoring traffic and enhancing urban safety.



Figure 6. Trained Sample Result Using YOLOv8

Instance Segmentation

In this section, bicycle lane segmentation model was the focus, which aimed to accurately identify and delineate bicycle lanes in various urban environments. The training process involved a carefully curated dataset containing diverse examples of bicycle lanes under different conditions, ensuring that the model could generalize well to real-world scenarios.

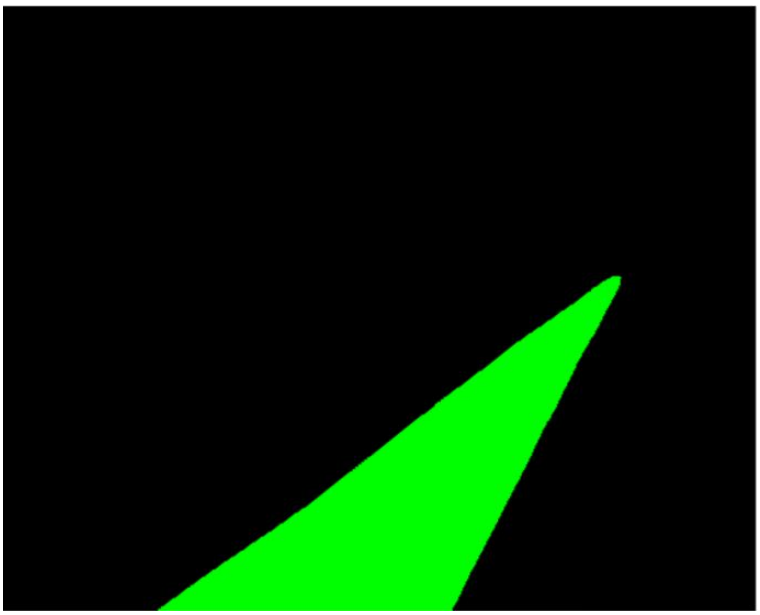


Figure 7. Trained Sample Result on Bicycle Lane Segmentation

It started by examining the training results, highlighting the model's performance in segmenting bicycle lanes. Followed by this, the Intersection over Union (IoU) was analyzed, a critical metric for evaluating the accuracy of the segmentation. Also, reviewed the validation loss over training epochs to assess how well the model learned from the data.

License Plate Recognition

In this section, license plate recognition was implemented using EasyOCR, a robust optical character recognition library and evaluated through Google Colab. For the dataset, some sample data were used to ensure a variety of license plates where most data contained similar images

with identical license plates. Pre-processing techniques were used to enhance recognition accuracy like denoise and simple grayscale.



(a) (b)

Figure 8. Applying Preprocessing in License Plate with (a) Image Denoising & (b) Grayscale Conversion

After applying the preprocessing techniques: image resizing, grayscale, denoise, adaptive scaling, and quality scaling – results showed a significant improvement in the accuracy of license plate recognition. Denoising helped reduce image noise, making the characters more distinct, while converting the images to grayscale enhanced contrast and clarity. This combination allowed EasyOCR to perform more efficiently, leading to higher character and plate recognition rates.

Motorcycle Intrusion on Bicycle Lanes

In this setup, the YOLOv8 model successfully spots motorcycles, marking those that enter bicycle lanes as "intrusion" with threshold value of 0.6. The U-Net helped create clear masks that outline important areas, making it easier to detect motorcycle intrusions and tell apart vehicles that are following the rules from those that are not. Custom code was used to automatically label motorcycles that entered the bicycle lane as violators. Each detected object was also given a unique ID to help track them across different frames.



Figure 9. Result for Detection of Motorcycle Intrusion and LPR

Figure 9 displays the result of the study using YOLOv8 to find motorcycle intrusions and Philippine license plates, along with U-Net for creating detailed masks. Overall, this model effectively detects motorcycle intrusions and recognizes license plates in bicycle lanes.

Model Performance Metrics

Assessing model performance was critical to determine the reliability of the YOLOv8s model for motorcycle intrusion detection and license plate recognition. By employing a variety of performance metrics, we can obtain a comprehensive understanding of how well the model function. Key metrics provide insights into the model's accuracy in identifying motorcycles and recognizing license plates. These metrics not only reflect the model's capability to correctly predict positive instances but also its ability to minimize false positives and negatives. This section will detail the performance results achieved by the optimized YOLOv8 model, highlighting improvements over baseline metrics and establishing a foundation for effective monitoring of motorcycle intrusions on bicycle lanes.

YOLOv8

The implementation of the YOLOv8s model for motorcycle intrusion detection and license plate recognition has yielded promising results, establishing a foundation for effective monitoring. YOLOv8 model demonstrated high performance across key metrics, including precision, recall, and mean Average Precision (mAP), as detailed in the Table below.

Table 5. Detection Evaluation Results

Metric	Baseline	Optimized	Improvement	% Change
Precision	0.8856	0.8815	-0.0041	-0.46%
Recall	0.8587	0.9014	+0.0427	+4.97%
F1-Score	0.8719	0.8913	+0.0194	+2.22%
mAP@50	0.9052	0.9318	+0.0266	+2.94%
mAP@50-95	0.6250	0.6388	+0.0138	+2.21%

This table presents the training results for the YOLOv8 model, showed a significant improvements in performance metrics. The optimized model achieved a precision of 88%, indicating that a substantial percentage of its positive predictions were correct, while recall increased to 90%, reflecting the model's enhanced ability to identify actual motorcycle instances. Moreover, F1-score improved to 89%, effectively balancing precision and recall for a comprehensive assessment of performance.

Furthermore, the mean Average Precision (mAP) at 50% IoU rose to 93%, demonstrating strong detection capabilities, while the mAP across IoU thresholds from 50% to 95% showed a slight increase to 64%. A confusion

matrix analysis revealed a significant number of true positive detections for motorcycles, underscoring the model's strong classification capabilities. The diverse dataset, which included various motorcycle instances in different environments, further enhanced the model's robustness. These metrics collectively highlight the model's reliability in accurately detecting and recognizing motorcycles and their license plates, thereby contributing to improved roadway safety and compliance.

After presenting the key metrics, study proceed to the model performed using different confidence curves. These curves helped to understand how well the model detected motorcycles and read license plates at varying confidence levels. Followed by the discussion about F1-Confidence Curve, Precision-Confidence Curve, Precision-Recall Curve, and Recall-Confidence Curve . Each of these curves showed how well the model balanced accuracy and the ability to find true positives, while also highlighting its strengths and areas that needed improvement.

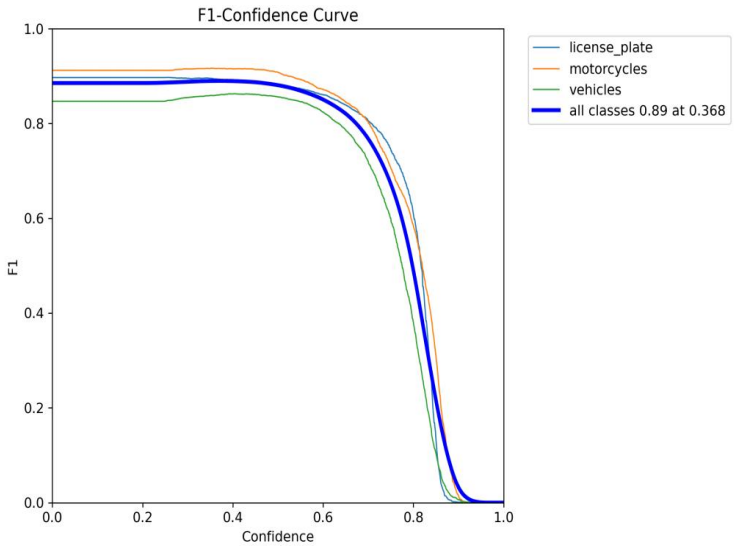


Figure 10. F1-Confidence

The F1-Confidence Curve illustrates the F1 score as a function of confidence levels for license plates, motorcycles, and vehicles, revealing that the model achieves a stable F1 score of approximately 0.89 at a confidence level of 0.368. This indicate that the model maintained a good balance between precision and recall, performed particularly well for motorcycles and license plates compared to vehicles, suggest and enhanced detection capabilities for these specific classes.

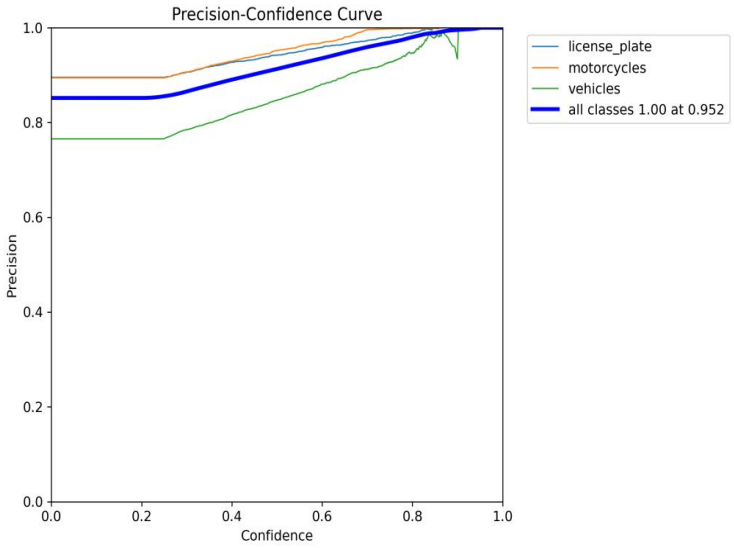


Figure 11. Precision-Confidence

The Precision-Confidence Curve displays how precision varies with confidence levels, showed that the model maintains high precision across the board, achieving a perfect score of 1.00 at a confidence level of 0.952 for all classes combined. This indicate that when the model is confident, it is highly accurate in its predictions, particularly for motorcycles and license plates, thereby minimized false positives and showcased the model's effectiveness in these categories.

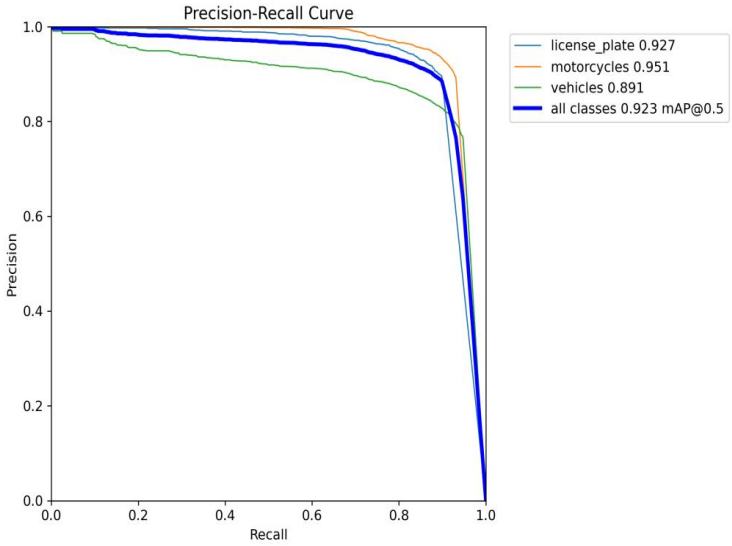


Figure 12. Precision-Recall

The Precision-Recall Curve depicts the trade-off between precision and recall at different thresholds, demonstrated high precision values (0.927 for license plates and 0.951 for motorcycles) while revealing some variability in recall. With an overall mean Average Precision (mAP) of approximately 0.923 at a threshold of 0.5, this graph indicate robust performance in detecting vehicles, highlighting the model's ability to identify true positives effectively while suggesting room for improvement in capturing all relevant instances.

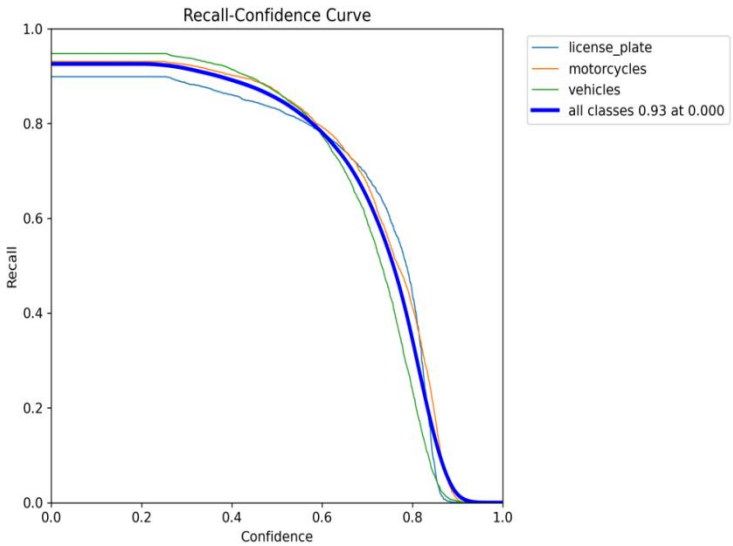
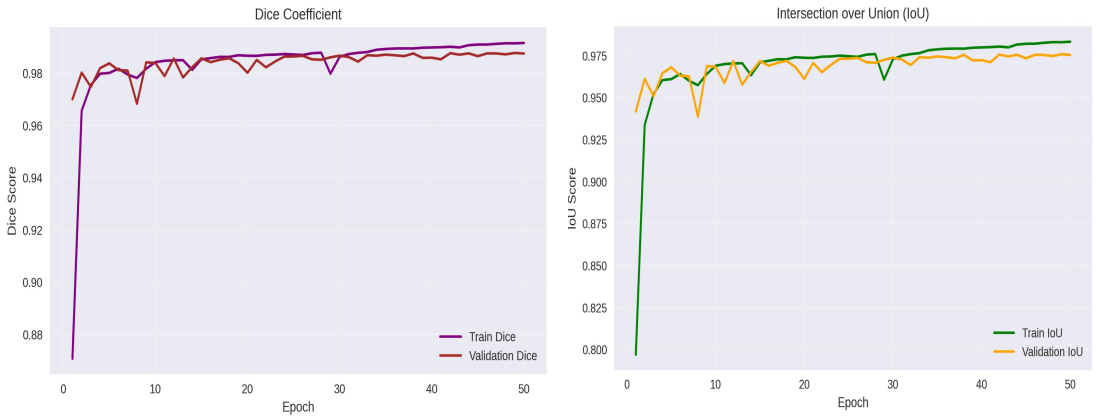


Figure 13. Recall-Confidence

The Recall-Confidence Curve shows how recall varies with confidence levels, indicated that the model maintains a high recall value of about 0.93 at lower confidence thresholds, effectively identifying a significant portion of actual instances. However, the gradual decline in recall as confidence increases suggest a trade-off where the model becomes more conservative in its positive predictions at higher thresholds, emphasized the need to balance precision and recall to capture more true positives without sacrificing accuracy.

U-Net with ResNet34

In bicycle lane segmentation, the model was evaluated based on the graph results of Dice Coefficient and Intersection over Union (IoU), which collectively measure the model's segmentation accuracy and overlap between predicted and actual regions. The evaluation began by examining the training results to highlight the model's effectiveness in segmenting bicycle lanes, followed by an analysis of the IoU as a key metric for assessing segmentation accuracy.



(a) (b)

Figure 14. Results in (a) Dice Coefficient and (b) Intersection over Union (IoU)

Figure 14 illustrates that the bicycle lane segmentation model performs exceptionally well, with a Dice coefficient stabilized around 0.98 and IoU scores around 0.975 for both training and validation. These metrics indicate

high accuracy and strong agreement between predicted and ground truth areas, confirming the model's robustness and reliability in segmenting bicycle lanes.

Consequently, the system used a U-Net architecture with a ResNet34 backbone for image processing, trained over 50 epochs with a batch size of 8. The training takes approximately 0.81 hours, utilized images resized to 512 × 512 pixels and a learning rate of 0.0001 for optimization.

Table 6. Model Configuration

Parameter	Description
Architecture	U-Net with ResNet34 backbone
Training Time	0.81 hours
Epochs	50
Batch Size	8
Image Size	512 × 512
Learning Rate	0.0001

Table 7 shows the model achieved a best validation IoU of 0.9718 and final Dice coefficients of 0.9836 for training and 0.9848 for validation, indicating high segmentation accuracy.

Table 7. Performance Summary

Metric	Training	Validation	Best/Final Value
Best Validation IoU			0.9718
Best Validation Loss			0.0080
Final IoU	0.9677	0.9701	
Final Dice Coefficient	0.9836	0.9848	

EasyOCR

The OCR model was evaluated using a total of 65 processed images, yielding 126 valid samples after data validation. Evaluation achieved an exact match accuracy of 61.11% (77 out of 126), indicate that more than half of the license plates were correctly recognized in their entirety. Additionally, the character accuracy reached 83.82%, demonstrate that most individual characters were accurately identified even when full plate matches were not achieved. In summary, EasyOCR demonstrated significant potential for license plate recognition; however, continuous evaluation and optimization remained essential to achieve greater reliability across diverse scenarios.

Table 8. EasyOCR Evaluation Results

License Plate	Ground Truth	OCR Result	OCR Confidence	Detection Confidence	Match
1	UWC142	UHC142	0.74	0.82	FALSE
2	5510XW	S510XW	0.49	0.86	FALSE
3	NFF5492	NFF5492	0.96	0.64	TRUE
4	NFF5492	NFF5492	0.99	0.72	TRUE
5	5492	5492	0.90	0.42	TRUE
6	UWC142	UHC142	0.29	0.82	FALSE
7	168XLJ	168XLJ	0.65	0.82	TRUE
8	158UBT	158UBT	1.00	0.70	TRUE
9	NGS7517	NGS7517	0.90	0.33	TRUE
10	DBI2913	DBI2913	0.95	0.79	TRUE
11	DBI2913	DBI2913	1.00	0.78	TRUE
12	DBI2913	DBI2913	0.81	0.82	TRUE
13	241XII	Z41XII	0.37	0.80	FALSE
14	NAI2435	NAI2435	0.71	0.44	TRUE
15	NIB1371	NIB1371	0.98	0.77	TRUE
16	DAL2225	DAL2225	1.00	0.81	TRUE
17	DAL2225	DAL2225	1.00	0.82	TRUE
18	390DBH	390DBH	0.76	0.86	TRUE
19	346XRM	346XRH	0.94	0.82	FALSE
20	NKC7692	NKC7692	0.97	0.80	TRUE

Table 8 presents a partial view of EasyOCR evaluation results. The original dataset contains a total of 128 rows, which were used in the analysis and computation of performance metrics. Evaluation result of the EasyOCR license plate recognition indicate a mixed performance, showcased both successful and unsuccessful recognitions. High OCR confidence scores, particularly those above 0.9, often correlate with accurate matches, suggesting that the model performs well under favorable conditions. However, lower confidence scores and detection highlight areas needed improvements, particularly regarding image quality and detection algorithms. Instances of false positives revealed the necessity for refining preprocessing techniques to enhance model accuracy.

Model Validation

This phase involved testing the developed models for vehicle detection, license plate recognition, and bicycle lane segmentation with separate validation data. It ensured accuracy and effectiveness while preventing overfitting, confirming reliability prior to final evaluation.

Vehicle and License Plate Detection

For this result, it has the decreasing losses, indicate that the model is improving in its ability to predict accurately over time.

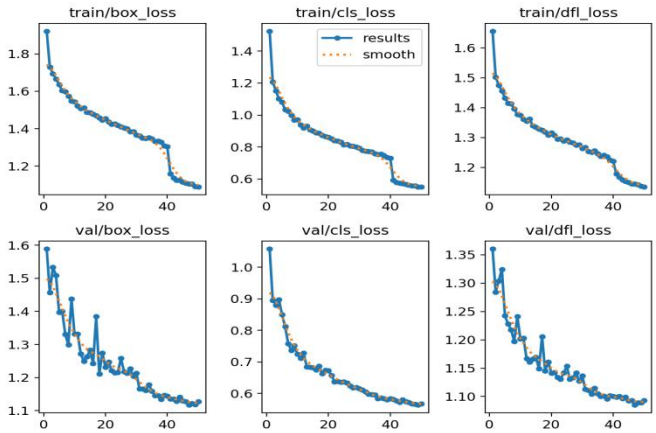


Figure 15. Validation Losses and Learning Rate During Training

Figure 15 displays training and validation loss metrics for a machine learning model. The top row showed the loss for bounding box predictions, classification, and a derived loss, all of which decrease over epochs, indicating improved model performance. Bottom row similarly illustrated validation losses, which also decline, suggest that the model was effective in learning and generalizing from the training data.

Figure 16 demonstrates how well the model distinguished between classes, achieved high accuracy for license plates (0.90) and motorcycles (0.93), while showing some confusion with vehicles and background. Indicate that even the model performed well overall, there is a need for further refinement to reduce misclassifications between vehicles and background classes.

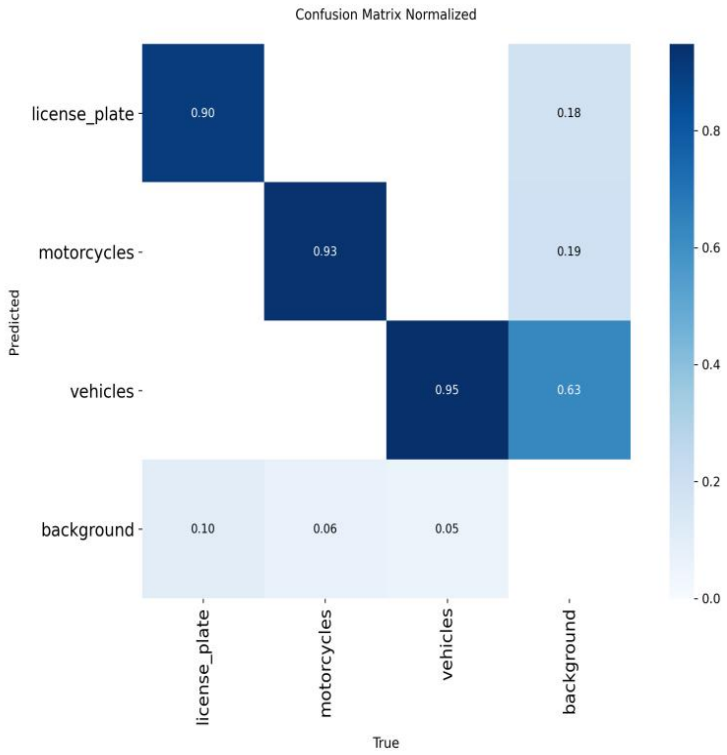


Figure 16. Confusion Matrix Normalized

Figure 17 visualizes the model's prediction performance across categories: license plates, motorcycles, vehicles, and background. It showed strong true positives, with the model accurately identifying 3,818 license plates and 3,450 motorcycles. However, there are notable misclassifications outside the diagonal, particularly with vehicles, where the model incorrectly classified 6 as vehicles and misidentified 4,237 vehicles in total. Additionally, 1,287 instances were wrongly classified as motorcycles. For background elements, 359 were mistaken for license plates, 236 as motorcycles, and 230 as vehicles. These findings indicate that while the model excel in recognizing license plates and motorcycles, it can still face significant challenges in accurately classifying vehicles, pointing to areas needing enhancement to improve overall accuracy.

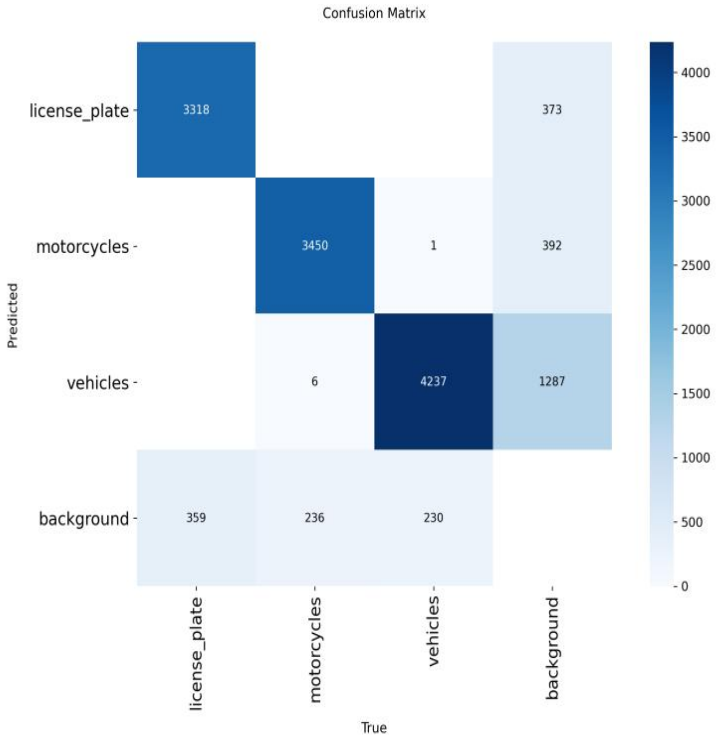


Figure 17. Confusion Matrix

Bicycle Lane Segmentation

The training loss decreased consistently, indicating effective learning, while the validation loss remained low, suggest that the model generalized well to unseen data.

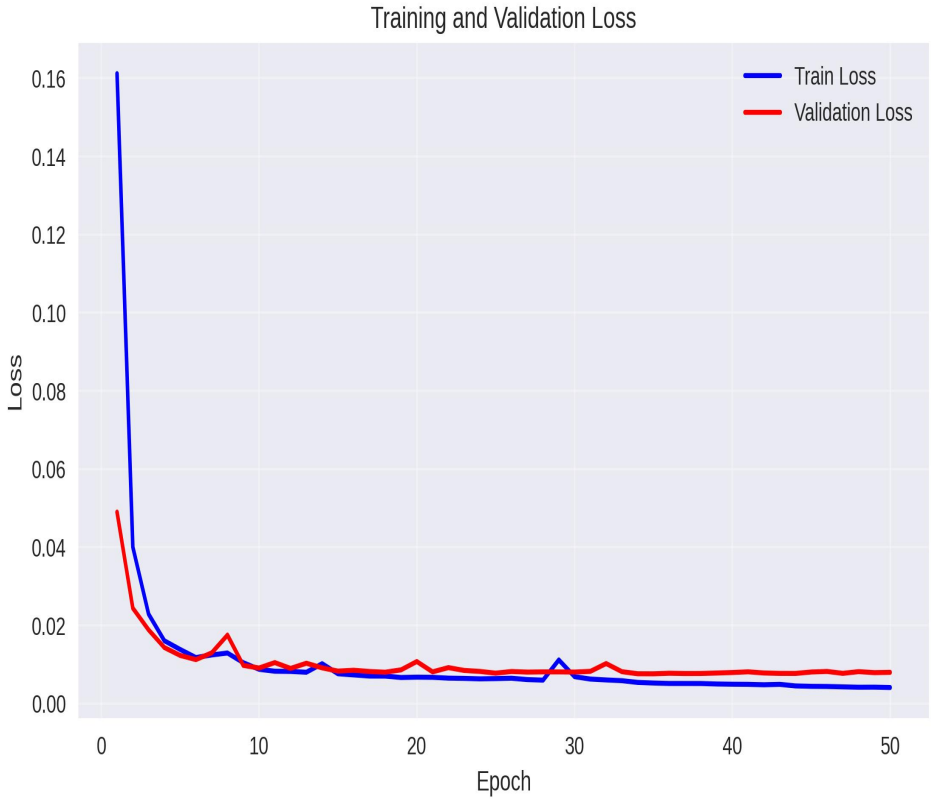


Figure 18. Training and Validation Loss

Motorcycle Intrusion Validation

In the validation process, researchers compared the model's performance with evaluation from expert at the Land Transportation Office (LTO) to check how accurately it identifies motorcycles in bicycle lanes. Fortunately, both the expert and the model found similar results. However, to strengthen the findings, the team compared the number of violations noted by the expert with those detected by the model in order for comparison to confirmed the model's accuracy and highlighted areas for improvement.

Table 9 summarizes the performance metrics of the model, showed a precision of 98.38%, recall of 97.74%, and an F1-score of 98.06%, indicate a strong detection capabilities. The mean average error (MAE) was 0.12, with a mean absolute percentage error (MAPE) of 3.09, provide insights into the model's overall accuracy and prediction reliability.

Table 9. Performance Metrics Results

Performance Metrics Summary	
Precision	98.38%
Recall	97.74%
F1-Score	98.06%
MAE	0.12
MAPE	3.09

Table 10 presents the results of a paired t-test, which is a statistical method used to determine whether there are significant differences between the means of two related groups. The test statistic obtained is 45.5a, representing a measure of the difference between the groups, although the p-value of 0.594 suggests that this difference is not statistically significant. Typically, a p-value below 0.05 is considered significant; however, in this case, the high p-value indicates that there is no meaningful difference between the two groups being compared.

Furthermore, the standard error of the mean difference is reported as 0.0348, which reflects the reliability and precision of this estimate. A smaller standard error indicates that the mean difference is stable and that repeated sampling would yield similar results.

Table 10. Statistical Test Results

Metric	Value
Statistic	45.5[a]
p-value	0.594
Number of Pairs(N)	100
Mean Difference	0.500
Standard Error Difference	0.0348

In this case, the W statistic is 0.473, and the p-value is less than 0.001. A low p-value where the differences do not follow a normal pattern.

Table 11. Test of Normality (Shapiro–Wilk)

			W	p
Expert Count	-	Model Count	0.473	<.001

This is important because it means we cannot use methods that assume normality, like the paired t-test, and instead should use non-parametric methods, such as the Wilcoxon signed-rank test. The Shapiro-Wilk test confirmed of using the right statistical methods for the data, which helped ensure that the findings about the model's ability to detect motorcycle intrusions were reliable.

For the descriptive summary, the findings indicate that both expert and model counts demonstrate a strong alignment in their performance, revealing that they are closely matched in terms of average ratings. The analysis highlights that the differences between the two groups are minimal, suggesting a consistent level of quality and effectiveness across both sets.

Table 12. Descriptive Summary

	N	Mean	Median	SD	SE
Expert Count	100	3.10	4.00	2.34	0.234
Model Count	100	3.08	4.00	2.31	0.231

Table 12 offers a detailed descriptive summary of the expert count and model count performance metrics. Both metrics have a sample size (N) of 100, made the results robust. The expert count has a mean of 3.10, indicating that experts tend to rate just above three, with a median of 4.00 suggesting that half of the expert ratings are four or higher. The standard deviation (SD) of 2.34 indicates moderate variability in the expert counts, while the standard error (SE) of 0.234 reflects the accuracy of the mean estimate. In contrast, the model count has a mean of 3.08, slightly lower than the expert count, but the median remains the same at 4.00, reinforcing consistency between the two metrics. The model count's standard deviation of 2.31 and standard error of 0.231 show similar variability and precision compared to the expert count.

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary

This study focused on the development and evaluation of deep learning models for the detection of motorcycle intrusions and license plate recognition on bicycle lanes. Utilizing the optimized YOLOv8 model, the research aimed to address the critical safety concerns posed by unauthorized motorcycle entries into designated bicycle lanes, thereby enhancing road safety for cyclists and pedestrians.

The research methodology involved the collection and annotation of a diverse dataset consisting of 1,832 raw images, which included motorcycles, bicycle laness, and license plates. Two primary models were developed: YOLOv8 for vehicle and license plate detection, and U-Net with ResNet34 for bicycle lane segmentation. The performance of these models was rigorously assessed through various metrics, including Precision, Recall, F1 Score, and Mean Average Precision (mAP) for detection, as well as the Dice coefficient and Intersection over Union (IoU) for segmentation accuracy.

The findings revealed that the YOLOv8 model achieved a precision of 88% and an mAP of 91%, indicating strong performance in accurately detecting motorcycles and their associated license plates. The segmentation model demonstrated an IoU of approximately 0.975, confirmed its reliability in accurately identifying bicycle lane boundaries. Additionally, EasyOCR was integrated for license plate recognition, achieving a character accuracy of 83.82%, highlighting its potential in accurately reading Philippine license plates. However, further optimization is necessary to improve its performance

in handling different plate formats and to accurately read entire Philippine license plates.

In the validation process, researchers compared the performance of the Yolov8 model against evaluations from experts at the Land Transportation Office (LTO) to assess how accurately it detects motorcycles in bicycle lanes. For the results, the performance metrics showed a precision of 98.38%, recall of 97.74%, and an F1-score of 98.06%. Finally, the Wilcoxon signed-rank test highlighted a no significant difference with a p-value of 0.594, confirmed the model's accuracy compared to expert assessments.

Conclusions

In conclusion, this study developed an optimized YOLOv8 model for detecting motorcycle intrusions and recognizing license plates on bicycle lanes to address cyclist and pedestrian safety. Using a dataset of 1,691 annotated images, YOLOv8 achieved 88% precision and 91% mAP, while the U-Net with ResNet34 model for lane segmentation attained an IoU of 0.975. EasyOCR integration for license plate recognition achieved 83.82% character accuracy, demonstrating strong performance for real-time monitoring.

The validation process demonstrated that the Yolov8 model performed effectively in detecting motorcycles in bicycle lanes, as evidenced by the analysis using 100 images with a precision of 94.74%, recall of 87.54%, and an F1-score of 91%. The performance metrics affirmed the model's strength and reliability. Furthermore, results of the Wilcoxon signed-rank test confirmed a no significant difference between the expert and model assessment with p-value of <0.594, indicate that the proposed model is reliable and produce results comparable to expert evaluation.

Recommendations

Based on the findings of this study, the following recommendations are proposed to enhance the performance of the deep learning models for motorcycle intrusion detection and license plate recognition:

1. **Expand the dataset and address challenging conditions.** Collect more diverse data from various road types, lighting, and weather conditions, including rain, fog, and occlusions, to enhance model robustness, reduce redundancy, and improve detection accuracy in complex environments.

2. **Explore advanced detection models**. Implement the latest versions of YOLO or other state-of-the-art object detection algorithms to improve detection accuracy, processing speed, and system efficiency.

3. **Include additional classes for better distinction.** Add more object classes to include other potential violators, not just motorcycles, such as cars, trucks, and other motor vehicles, along with non-violators like persons and bicycles, to enhance detection accuracy and reduce confusion in classification.

4. **Explore various camera angles and placements.** Future studies should focus on capturing data from multiple CCTV angles and camera positions to enhance both intrusion detection and license plate recognition. By exploring different viewpoints, the model can improve detection accuracy and enable clearer reading of license plates.

5. **Train the model for license plate recognition.** Future studies are encouraged to train a dedicated model using a larger and more diverse set of license plate images with varying angles, lighting conditions, and font styles. This approach can enhance model's ability to recognize and read full plates more accurately.

REFERENCES

Aboyomi, D., & Daniel, C. (2023). A Comparative Analysis of Modern Object Detection Algorithms: YOLO vs. SSD vs. Faster R-CNN. ITEJ (Information Technology Engineering Journals), 8(2), 96 - 106. https://doi.org/10.24235/itej.v8i2.123

Ahmad, I. S., Boufama, B., Habashi, P., Anderson, W., & Elamsy, T. (2015, December). Automatic license plate recognition: A comparative study. In 2015 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT) (pp. 635-640). IEEE.

Ahmed, M. J., Sarfraz, M., Zidouri, A., & Al-Khatib, W. G. (2003, December). License plate recognition system. In 10th IEEE International Conference on Electronics, Circuits and Systems, 2003. ICECS 2003. Proceedings of the 2003 (Vol. 2, pp. 898-901). IEEE.

Ahmed, S., Huda, M. N., Rajbhandari, S., Saha, C., Elshaw, M., & Kanarachos, S. (2019). Pedestrian and cyclist detection and intent estimation for autonomous vehicles: A survey. Applied Sciences, 9(11), 2335.

Antwi, R. B., Lawson, P. L., Kimollo, M., Ozguven, E. E., Moses, R., Dulebenets, M. A., & Sando, T. (2024). Automated Detection of Pedestrian and Bicycle Lanes from High-Resolution Aerial Images by Integrating Image Processing and Artificial Intelligence (AI) Techniques.

Azimjonov, J., Özmen, A., & Varan, M. (2023). A vision-based real-time traffic flow monitoring system for road intersections. Multimedia tools and applications, 1–20. Advance online publication. https://doi.org/10.1007/s11042-023-14418-w

Azzakhnini, M., Saidi, H., Azough, A., Tairi, H., & Qjidaa, H. (2025). LAVID: A Lightweight and Autonomous Smart Camera System for Urban Violence Detection and Geolocation. Computers, 14(4), 140. https://doi.org/10.3390/computers14040140

Babu, A. M., Akhil, B., & Pochampally, N. K. (2023). Smart Cities and Intelligent Transport Systems. International Journal of Applied and Structural Mechanics, 31, 22–25. https://doi.org/10.55529/ijasm.31.22.25

Camba, A. L. A., Dimayuga, D. G., & Doroy, M. R. A. (2017). Bicycle commuting practices in Marikina City, Philippines: A case study. Transportation Science Society of the Philippines. https://ncts.upd.edu.ph/tssp/wp- content/uploads/2017/07/TSSP2017-01-Camba-Dimayuga-and- Doroy.pdf

Chang, W. J., Chen, L. B., & Su, K. Y. (2019). DeepCrash: A deep learning-based internet of vehicles system for head-on and single-vehicle accident detection with emergency notification. IEEE Access, 7, 148163- 148175.

Chandra, N., & Panda, S. P. (2021, August). A human intruder detection system for restricted sensitive areas. In 2021 2nd International Conference on Range Technology (ICORT) (pp. 1-4). IEEE.

Cruz, G. G. L., Litonjua, A., San Juan, A. N. P., Libatique, N. J., Tan, M. I. L., & Honrado, J. L. E. (2022, September). Motorcycle and Vehicle Detection for Applications in Road Safety and Traffic Monitoring Systems. In 2022 IEEE Global Humanitarian Technology Conference (GHTC) (pp. 102-105). IEEE.

Daily Tribune. (2024, July 16). Daily accidents and elusive progress. The Philippine Star.

Dequito, C. J. M., Dichaves, I. J. L., Juan, R. J. G., Minaga, M. Y. K. T., Ilao, J. P., Cordel, I. M., & Del Gallego, N. P. A. (2021, May). Vision-based bicycle and motorcycle detection using a yolo-based network. In Journal of Physics: Conference Series (Vol. 1922, No. 1, p. 012003). IOP Publishing.

Devi, S. C., Reddy, G. V., Aakarsh, Y., & Gowtham, B. (2022). License plate detection of motorcyclists without helmets. In Computational Intelligence and Data Analytics: Proceedings of ICCIDA 2022 (pp. 295-301). Singapore: Springer Nature Singapore.

Dhedhi, B., Datar, P., Chiplunkar, A., Jain, K., Rangarajan, A., & Kundargi, J. (2018). Automatic License Plate Recognition Using Deep Learning (pp. 46–58). Springer, Singapore.

Dhillon, A., & Verma, G. K. (2020). Convolutional neural network: a review of models, methodologies and applications to object detection. Progress in Artificial Intelligence, 9(2), 85-112.

Du, S., Ibrahim, M., Shehata, M., & Badawy, W. (2013). Automatic License Plate Recognition (ALPR): A State-of-the-Art Review. IEEE Transactions on Circuits and Systems for Video Technology, 23(2), 311– 325.

Duan, T. D., Du, T. H., Phuoc, T. V., & Hoang, N. V. (2005, February). Building an automatic vehicle license plate recognition system. In Proc. Int. Conf. Comput. Sci. RIVF (Vol. 1, pp. 59-63).

Egon, K., & KARL, L. (2023). Transfer Learning and Pretrained Models.

Espinosa, J. E., Velastin, S. A., & Branch, J. W. (2018, May). Motorcycle detection and classification in urban Scenarios using a model based on Faster R-CNN. In 9th International Conference on Pattern Recognition Systems (ICPRS 2018) (p. 16). Stevenage UK: IET.

Espinosa-Oviedo, J. E., Velastín, S. A., & Branch-Bedoya, J. W. (2019). EspiNet V2: a region based deep learning model for detecting motorcycles in urban scenarios. Dyna, 86(211),1 317-326.

Garcia-Pajuelo, J., & Paiva-Peredo, E. Comparison and evaluation of YOLO models for vehicle detection on bicycle paths. Int J Artif Intell ISSN, 2252(8938), 3635.

GeeksforGeeks. (2023). Convolutional Neural Network (CNN) architectures. GeeksforGeeks.

GeeksforGeeks. (2025, July 23). Convolutional Neural Network (CNN) architectures. GeeksforGeeks. https://www.geeksforgeeks.org/machine-learning/convolutional-neural-network-cnn-architectures/

Gholamhosseinian, A., & Seitz, J. (2021). Vehicle classification in intelligent transportsystems: An overview, methods and software perspective. IEEE Open Journal of Intelligent Transportation Systems, 2, 173-194.

Gorak, T., & Reddy, C. K. K. (2024). Helmet and Number Plate Detection Using YOLOV5 and YOLOV8. International Journal of Interpreting Enigma Engineers (IJIEE), 1(4), 14-22.

Haghighat, A. K., Ravichandra-Mouli, V., Chakraborty, P., Esfandiari, Y., Arabi, S., & Sharma, A. (2020). Applications of deep learning in intelligent transportation systems. Journal of Big Data Analytics in Transportation, 2, 115-145.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In Proceedings of the IEEE conference on computer vision and pattern2 recognition (pp. 770-778).

Hernández-Díaz, N., Peñaloza, Y. C., Rios, Y. Y., Martinez-Santos, J. C., & Puertas, E. (2024). A computer vision system for detecting motorcycle violations in pedestrian zones. Multimedia Tools and Applications, 1-24.

Hoang, T. M., Nam, S. H., & Park, K. R. (2019). Enhanced detection and recognition of road markings based on adaptive region of interest and deep learning. IEEE Access, 7, 109817-109832.

Hotkar, O., Radhakrishnan, P., Singh, A., Jhamnani, N., & Bidwe, R. V. (2023). U-Net and YOLO: AIML Models for Lane and Object Detection in Real-Time. https://doi.org/10.1145/3607947.3608049

Indira, K., Mohan, K. V., & Nikhilashwary, T. (2019). Automatic license plate recognition. In Recent Trends in Signal and Image Processing: ISSIP 2017 (pp. 67-77). Springer Singapore.

Jayakumar, D., Saikia, S., Roshni, S. Y., Peddakrishna, S., Kollem, S., Naresh, M., & Krishnaiah, M. (2024). Lane Detection with U-Net Optimization through Hyperparameter Tuning and Cross Validation. 1– 6.

Jørgensen, H. (2017). Automatic license plate recognition using deep learning techniques (Master's thesis, NTNU).

Joshi, R. C., Singh, A. G., Joshi, M., & Mathur, S. (2019). A low cost and computationally efficient approach for occlusion handling in video surveillance systems.

Kaur, P., Kumar, Y., Ahmed, S., Alhumam, A., Singla, R., & Ijaz, M. F. (2022). Automatic License Plate Recognition System for Vehicles Using a CNN. Computers, Materials & Continua, 71(1).

Khan, S. A., & Jiang, R. (2019). Video surveillance-based intelligent traffic management in smart cities. Intelligent Video Surveillance, 19.

Khan, S. W., Hafeez, Q., Khalid, M. I., Alroobaea, R., Hussain, S., Iqbal, J., Almotiri, J., & Ullah, S. S. (2022). Anomaly Detection in Traffic Surveillance Videos Using Deep Learning. Sensors, 22(17), 6563. https://doi.org/10.3390/s22176563

Krishna, N. M., Reddy, R. Y., Reddy, M. S. C., Madhav, K. P., & Sudham, G. (2021, September). Object detection and tracking using YOLO. In 2021 Third International Conference on Inventive Research in Computing Applications (ICIRCA) (pp. 1-7). IEEE.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. Advances in neural information processing systems,3 25.

Kulchandani, J. S., & Dangarwala, K. J. (2015). Moving object detection: Review of recent research trends. International Conference on Pervasive Computing, 1–5.

Lan, J., Jiang, Y., Fan, G., Yu, D., & Zhang, Q. (2016). Real-time automatic obstacle detection method for traffic surveillance in urban traffic. Journal of Signal Processing Systems, 82, 357-371.

Laroca, R., Severo, E., Zanlorensi, L. A., Oliveira, L. S., Gonçalves, G. R., Schwartz, W. R., & Menotti, D. (2018, July). A robust real-time automatic license plate recognition based on the YOLO detector. In 2018 international joint conference on neural networks (ijcnn) (pp. 1-10). IEEE.

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. Proceedings of the IEEE, 86(11), 2278-2324.

LETHO, K., & Potter, K. (2023). Knowledge Transfer in Machine Learning, Adaptive Learning with Pretrained Models.

Liu, B., Yu, L., Che, C., Lin, Q., Hu, H., & Zhao, X. (2023). Integration and performance analysis of artificial intelligence and computer vision based on deep learning algorithms. arXiv preprint arXiv:2312.12872.

Lu, G. (2015). A lane detection, tracking and recognition system for smart vehicles (Doctoral dissertation, Université d'Ottawa/University of Ottawa).

Luo, X., Wang, Y., Cai, B., & Li, Z. (2021). Moving object detection in traffic surveillance video: new MOD-AT method based on adaptive threshold. ISPRS International Journal of Geo-Information, 10(11), 742.

Lv, Z., Zhang, S., & Xiu, W. (2020). Solving the security problem of intelligent transportation system with deep learning. IEEE Transactions on Intelligent Transportation Systems, 22(7), 4281-4290.

Mahmoud, A., & Ahmed, S. (2024). Deep Learning for Computer Vision: Innovations in Image Recognition and Processing Techniques. CyberSystem Journal, 1(1), 23-32.

Man-Woo Park, Jung Kim, Young-Joo Lee, Jinwoo Park, and Wonho Suh. 2017. Vision-based surveillance system for monitoring traffic conditions. Multimedia Tools Appl. 76, 23 (December 2017), 25343–25367. https://doi.org/10.1007/s11042-017-4521-4

Masula, J., Tingga, R., Salido, J. A., Española, J. C., & Gillesania, G. K. (2024). Motorcycle Recognition System Using Convolutional Neural Network. Journal of Innovative Technology Convergence, 6(3).

Maurya, A. R. (2023). Deep Learning Based Automatic Vehicle License Plate Recognition System for Enhanced Vehicle Identification. International Journal on Recent and Innovation Trends in Computing and Communication.

Metropolitan Manila Development Authority. (2024, April 9). MMDA logs 17.3 % increase in motorcycle-related crashes in 2023 [News release]. Inquirer.net. Inquirer.net

Mukhija, P., Dahiya, P. K., & Priyanka, P. (2021, July). Challenges in automatic license plate recognition system: An indian scenario. In 2021 Fourth international conference on computational intelligence and communication technologies (CCICT) (pp. 255-259). IEEE.

Nong, M. A., Yusof, J., Osman, R., Sidek, R., & Sabil, S. (2020). Motorcycle Plated Recognition Based on FPGA. Computational Intelligence.

Ouardirhi, Z., Mahmoudi, S. A., & Zbakh, M. (2024). Enhancing Object Detection in Smart Video Surveillance: A Survey of Occlusion-Handling Approaches. Electronics, 13(3), 541. https://doi.org/10.3390/electronics13030541

Padalia, D. (2023). Detection and number plate recognition of non-helmeted motorcyclists using YOLO. Authorea Preprints.

Patil, S. S., Patil, S. H., Pawar, A. M., Bewoor, M. S., Kadam, A. K., Patkar, U. C., ... & Sharma, S. (2023, July). Vehicle number plate detection using yolov8 and easyocr. In 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT) (pp. 1-4). IEEE.

Patil, S., Waghule, S., Waje, S., Pawar, P., & Domb, S. (2024). Efficient object detection with YOLO: a comprehensive guide. Int J Adv Res Sci Communi Technol, 2022, 519-31.

Philippine Statistics Authority. (2024). “As of 2023, official data indicate that there are 8.5 million registered motorcycles in the country.” Senate of the Philippines+2Philstar+2

Rashid, N. U., Mithun, N. C., Joy, B. R., & Rahman, S. M. (2010, December). Detection and classification of vehicles from a video using time-spatial image. In International Conference on Electrical & Computer Engineering (ICECE 2010) (pp. 502-505). IEEE.

Ravichandran, K. S. (2025). Estimation of automatic license plate recognition using deep learning algorithms. Spectrum of Decision Making and Applications, 2(1), 100-119.

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You only look once: Unified, real-time object detection. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 77).

Statista. (n.d.). Number of registered private motorcycles and tricycles in the Philippines from 2016 to 2022 [Graph]. Statista. https://www.statista.com/statistics/708054/number-of-registered-private-motorcycles-and-tricycles-in-the-philippines

Salsabila, N. (2024). Enhancing Automated Vehicle License Plate Recognition with YOLOv8 and EasyOCR. Journal of Information Systems and Informatics, 6(3).

Sohan, M., Sai Ram, T., & Rami Reddy, C. V. (2024). A review on yolov8 and its advancements. In International Conference on Data Intelligence and Cognitive Informatics (pp. 529-545). Springer, Singapore.

Tahir, N. U. A., Long, Z., Zhang, Z., Asim, M., & ELAffendi, M. (2024). PVswin-YOLOv8s: UAV-Based Pedestrian and Vehicle Detection for Traffic Management in Smart Cities Using Improved YOLOv8. Drones, 8(3), 84. https://doi.org/10.3390/drones8030084

Vedhaviyassh, D. R., Sudhan, R., Saranya, G., Safa, M., & Arun, D. (2022, December). Comparative analysis of easyocr and tesseractocr for automatic license plate recognition using deep learning algorithm. In 2022 6th International Conference on Electronics, Communication and Aerospace Technology(pp. 966-971). IEEE.

Weihong, W., & Jiaoyang, T. (2020). Research on license plate recognition algorithms based on deep learning in complex environment. *IEEE Access*, *8*, 91661-91675.

World Health Organization. (2023). Road traffic injuries. Retrieved from https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries

APPENDICES

APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

"UTILIZATION OF DEEP LEARNING: OPTIMIZED YOLOV8 FOR DETECTION OF MOTORCYCLE INTRUSION AND LICENSE PLATE RECOGNITION ON BICYCLE LANES"

DAREL T. ALZARAZ, CHRIS F. BALOGO, RODA C. MENDEZ have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

JEFFERSON L. LERIOS
Thesis Project Adviser
October 30, 2025

APPENDIX B

Certificate of Acceptance

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF ACCEPTANCE

This is to certify that the data collection for the research project entitled **"UTILIZATION OF DEEP LEARNING: OPTIMIZED YOLOV8 FOR DETECTION OF MOTORCYCLE INTRUSION AND LICENSE PLATE RECOGNITION ON BICYCLE LANES"** has been conducted successfully in Cubao, Quezon City.

The data collection was carried out under the guidance and approval of our professor, Prof. Jefferson L. Lerios, who provided valuable insights and oversight throughout the process.

We affirm that all ethical considerations were observed, and the data was collected responsibly and in accordance with the standards set by our institution.

Date: August 20, 2025
Location: Cubao, Quezon City

Participants:

Alcaraz, Darel T.
Balogo, Chris F.
Mendez, Roda C.

Professor's Signature:

Jefferson L. Lerios

Prof. Jefferson L. Lerios
Thesis Adviser
College of Computer Studies
Laguna State Polytechnic University

APPENDIX C

Relevant Source Code

Optimized YOLOv8 Training Process

```
import os
from ultralytics import YOLO
import torch

class OptimizedTrainingConfig:
    # MODEL & TRAINING
    MODEL = "yolov8s.pt"
    EPOCHS = 50
    IMGSZ = 960  # Higher resolution than baseline (640px)
    BATCH = 8

    # OPTIMIZER (AdamW vs SGD in baseline)
    OPTIMIZER = "AdamW"
    LR0 = 0.001
    WEIGHT_DECAY = 0.0001

    # ADVANCED AUGMENTATION
    MIXUP = 0.5
    COPY_PASTE = 0.5
    MOSAIC = 1.0
    FLIPLR = 0.5

    # REGULARIZATION
    LABEL_SMOOTHING = 0.05
    PATIENCE = 25

def train_optimized_model():
    """Main training function with optimized parameters"""
    cfg = OptimizedTrainingConfig()
    model = YOLO(cfg.MODEL)

    results = model.train(
        data="dataset/data.yaml",
        epochs=cfg.EPOCHS,
        imgsz=cfg.IMGSZ,
        batch=cfg.BATCH,
        optimizer=cfg.OPTIMIZER,
        lr0=cfg.LR0,
        weight_decay=cfg.WEIGHT_DECAY,
        mixup=cfg.MIXUP,
        copy_paste=cfg.COPY_PASTE,
        mosaic=cfg.MOSAIC,
        fliplr=cfg.FLIPLR,
        label_smoothing=cfg.LABEL_SMOOTHING,
        patience=cfg.PATIENCE,
        cos_lr=True  # Cosine learning rate
    )
    return model, results
```

1. BASELINE YOLOv8 TRAINING CONFIGURATION

```
from ultralytics import YOLO

class BaselineTrainingConfig:
    # STANDARD YOLOv8 PARAMETERS (No optimizations)
    MODEL = 'yolov8s.pt'
    EPOCHS = 50
    IMG_SIZE = 640  # Standard resolution
    BATCH_SIZE = 8
    WORKERS = 4

def train_baseline_model():
    """Baseline training using YOLOv8 default parameters"""
    cfg = BaselineTrainingConfig()
    model = YOLO(cfg.MODEL)

    # Training with minimal parameters (YOLOv8 defaults)
    results = model.train(
        data="dataset/data.yaml",
        epochs=cfg.EPOCHS,
        imgsz=cfg.IMG_SIZE,
        batch=cfg.BATCH_SIZE,
        workers=cfg.WORKERS,
        # No custom optimizers, augmentation, or regularization
    )
    return model, results
```

2. BIKE LANE SEGMENTATION MODEL

```
import segmentation_models_pytorch as smp
import torch.nn as nn

class BikeLaneSegmenter:
    def __init__(self, img_size=512, num_classes=2):
        self.config = {
            'architecture': 'U-Net + ResNet34',
            'image_size': img_size,
            'classes': ['background', 'bike_lane'],
            'encoder_weights': 'imagenet',
            'activation': None
        }
    def create_model(self):
        """Initialize U-Net with ResNet34 backbone"""
        model = smp.Unet(
            encoder_name="resnet34",
            encoder_weights="imagenet",
            in_channels=3,
            classes=2,  # background + bike lane
            activation=None
        )
        return model
```

```
class TrainingConfig:
    """Bike Lane Segmentation Training Parameters"""
    # Training
    EPOCHS = 30
    BATCH_SIZE = 8
    LEARNING_RATE = 1e-4
    IMG_SIZE = 512

    # Loss & Optimization
    LOSS_WEIGHTS = [1.0, 2.0]  # Higher weight for bike lane class
    OPTIMIZER = "AdamW"
    # Data Augmentation
    AUGMENTATION = [
        'Horizontal Flip',
        'Random Brightness/Contrast',
        'Gaussian Noise',
        'Blur'
    ]
```

3. Integration of Multiple Computer Vision Models with Real-time Processing

```
class CompleteIntrusionSystem:
    def __init__(self):
        # Core Models
        self.vehicle_detector = YOLO('yolov8s.pt')  # Custom-trained object detection
        self.bike_lane_segmenter = BikeLaneSegmentationModel()  # U-Net + ResNet34
        self.license_reader = FixedAdvancedOCR()  # EasyOCR integration

        # System Modules
        self.object_tracker = SimplifiedObjectTracker()
        self.license_manager = LicensePlateManager()
        self.alert_system = AlertGenerator()
        # Configuration
        self.config = SystemConfig()

    def process_frame(self, frame):
        """
        Complete processing pipeline for each frame
        Integrates multiple computer vision models
        """
         - OBJECT DETECTION PHASE
        detections = self.vehicle_detector(
            frame,
            conf=self.config.DETECTION_CONFIDENCE
        )
        # Extract different object types
        motorcycles = [d for d in detections if self.is_motorcycle(d)]
        license_plates = [d for d in detections if self.is_license_plate(d)]
        vehicles = [d for d in detections if self.is_vehicle(d)]

        - BIKE LANE SEGMENTATION PHASE
        bike_lane_mask = self.bike_lane_segmenter.predict(frame)
```

```
    - INTRUSION DETECTION PHASE
    intrusions = []
    for motorcycle in motorcycles:
        if self.detect_intrusion(motorcycle, bike_lane_mask):
            intrusion_data = self.create_intrusion_record(motorcycle)
            intrusions.append(intrusion_data)

    - LICENSE PLATE PROCESSING PHASE
    license_data = []
    for intrusion in intrusions:
        license_info = self.process_license_plate(
            frame, intrusion, license_plates
        )
        license_data.append(license_info)

        # Update license manager with best reading
        self.license_manager.update_license_reading(
            intrusion['permanent_id'],
            license_info['text'],
            license_info['confidence']
        )

    # 5. ALERT GENERATION PHASE
    alerts = self.alert_system.generate_alerts(intrusions, license_data)

    # 6. VISUALIZATION PHASE
    result_frame = self.visualize_detections(
        frame, detections, intrusions, license_data, alerts
    )

    return {
        'processed_frame': result_frame,
        'detections': detections,
        'intrusions': intrusions,
        'license_data': license_data,
        'alerts': alerts
    }

def detect_intrusion(self, motorcycle, bike_lane_mask):
    """Check if motorcycle overlaps with bike lane"""
    bbox = motorcycle['bbox']
    motorcycle_mask = self.create_bbox_mask(bbox, bike_lane_mask.shape)

    overlap = np.logical_and(motorcycle_mask, bike_lane_mask)
    overlap_ratio = np.sum(overlap) / np.sum(motorcycle_mask)

    return overlap_ratio >= self.config.INTRUSION_THRESHOLD

def process_license_plate(self, frame, intrusion, license_plates):
    """Extract license plate text using OCR"""
    motorcycle_bbox = intrusion['detection']['bbox']
```

```
        # Find best matching license plate
        best_license = self.find_associated_license_plate(
            motorcycle_bbox, license_plates
        )

        if best_license:
            # Extract text using OCR
            license_bbox = best_license['bbox']
            text, confidence = self.license_reader.read_license_plate(
                frame, license_bbox
            )

            return {
                'text': text,
                'confidence': confidence,
                'bbox': license_bbox,
                'associated_motorcycle': intrusion['permanent_id']
            }

        return {'text': 'No License Detected', 'confidence': 0.0}

class SystemConfig:
    """Complete system configuration parameters"""
    # Detection Parameters
    DETECTION_CONFIDENCE = 0.6
    INTRUSION_THRESHOLD = 0.1
    LICENSE_CONFIDENCE_THRESHOLD = 0.05

    # Processing Parameters
    IMAGE_SIZE = 640
    BATCH_SIZE = 8
    TARGET_FPS = 30

    # Alert Parameters
    ALERT_COOLDOWN = 20
    VIOLATION_THRESHOLD = 2

    # Visualization Parameters
    COLORS = {
        'motorcycle': (0, 255, 0),
        'intrusion': (0, 0, 255),
        'license_plate': (0, 255, 255),
        'vehicle': (255, 0, 0)
    }

class GradioInterface:
    """Web interface for system interaction"""
    def __init__(self, detection_system):
        self.system = detection_system

    def create_interface(self):
        """Create Gradio web interface"""
```

```
    with gr.Blocks(title="Motorcycle Intrusion Detection") as interface:

        # Image Processing Tab
        with gr.Tab("Image Processing"):
            self.create_image_interface()

        # Video Processing Tab
        with gr.Tab("Video Processing"):
            self.create_video_interface()

        # Results Tab
        with gr.Tab("Results & Analytics"):
            self.create_analytics_interface()

    return interface

def process_image_input(self, image):
    """Process single image through complete system"""
    # Convert and process image
    frame = self.pil_to_cv2(image)
    results = self.system.process_frame(frame)

    # Generate summary
    summary = self.generate_summary(results)

    return results['processed_frame'], summary

def process_video_input(self, video_path):
    """Process video file through complete system"""
    cap = cv2.VideoCapture(video_path)
    output_path = "processed_video.mp4"

    # Process each frame
    while True:
        ret, frame = cap.read()
        if not ret:
            break

        results = self.system.process_frame(frame)
        self.write_frame(output_path, results['processed_frame'])

    # Generate comprehensive report
    report = self.generate_video_report()

    return output_path, report
```

APPENDIX D

Evaluation Process and Tool



Figure D1. Dataset Verification

Researchers collaborate with industry expert to verify the data utilized in their thesis, ensuring its accuracy and efficiency. This step is crucial for reinforcing the credibility of findings and enhancing the overall result of the study.

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

UTILIZATION OF DEEP LEARNING: OPTIMIZED YOLOV8 FOR DETECTION OF MOTORCYCLE INTRUSION AND LICENSE PLATE RECOGNITION ON BICYCLE LANES

Darel T. Alcaraz, Chris F. Balogo, Roda C. Mendez
Researchers

We kindly request your permission to validate the dataset for use in our thesis, "Utilization Of Deep Learning: Optimized Yolov8 For Detection Of Motorcycle Intrusion And License Plate Recognition On Bicycle Lanes." This validation is crucial as it will enhance the reliability and credibility of our research findings.

The validation will ensure that:

1. The dataset accurately represents the scenarios of motorcycle intrusions and license plate visibility on bicycle lanes.

2. The simulation model correctly implements and visualizes the performance of the YOLOv8 algorithm under various conditions.

Validation Procedure

1. Dateset Creation: The dataset was manually created by the researchers through field data collection in Cubao, particularly focusing on bicycle lanes where motorcycle intrusions frequently occur. We recorded videos of these lanes and subsequently extracted frames from the footage. These frames were then uploaded to Roboflow for annotation and preprocessing, resulting in our final dataset.

2. Verification: The dataset will be cross-verified for:

- Accuracy of labeled data (motorcycles and license plates).
- Consistency in image quality and resolution.
- Diversity of scenarios (lighting, angles, weather conditions).

3.Model Testing: The YOLOv8 model will be tested to ensure:

- Correct detection rates of motorcycles and license plates.
- Performance metrics such as precision, recall, and F1 score.
- Visualization clarity of detection results.

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

CERTIFICATION OF VALIDATION

This is to certify that I, JANICE C. SOMBILLO, MIT, have validated the dataset and model used in the study titled "Utilization of Deep Learning: Optimized YOLOv8 for Detection of Motorcycle Intrusion and License Plate Recognition on Bicycle Lanes." I confirm that the dataset accurately represents the scenarios studied and is suitable for use in the research.

Issued this 04 day of NOVEMBER, 2025, at Los banos, Laguna

Signed by:

JANICE C. SOMBILLO
Signature over printed name

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

Validation Criteria
(Put a check in the box)

Criteria	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)
Accuracy of labeled motorcycles and license plates		✓			
Consistency of image quality and resolution	✓				
Diversity of scenarios represented in the dataset		✓			
Correctness of detection rates and performance metrics		✓			
Clarity of visualization of detection results		✓			
Overall usability and functionality of the dataset and model	✓				

Comments and Suggestions:

Overall Rating

☑ Excellent – Dataset is accurate and valid for research use.
☐ Very Good – Dataset is valid with minor revisions.
☐ Good – Dataset requires moderate revisions.
☐ Fair – Dataset needs major revisions.
☐ Poor – Dataset invalid; reconstruction required.

z

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

UTILIZATION OF DEEP LEARNING: OPTIMIZED YOLOV8 FOR DETECTION OF MOTORCYCLE INTRUSION AND LICENSE PLATE RECOGNITION ON BICYCLE LANES

Darel T. Alcaraz, Chris F. Balogo, Roda C. Mendez
Researchers

We sincerely appreciate your time and expertise in participating as a validator for this study. Kindly review each image carefully and indicate whether a motorcycle is present or not present on the designated bicycle lane. Your professional assessment will play a vital role in verifying the accuracy of the developed model and ensuring the reliability of the study's data. Please be assured that all information provided will be used solely for academic purposes and handled with the utmost confidentiality.

Instruction:
On Bicycle Lane, indicate **1** if a **motorcycle is present and provide number of motorcycle intrusion** and **0** if **no motorcycle is present**.

Sample Data	*LTO Expert*	
	1=Motorcycle Present 0= Not Present	*Number of Motorcycle Intrusion*
test1.jpg	1	5
test2.jpg	0	0
test3.jpg	1	5
test4.jpg	1	5
test5.jpg	0	0
test6.jpg	1	6
test7.jpg	1	6
test8.jpg	1	6
test9.jpg	1	6
test10.jpg	1	6
test11.jpg	0	0
test12.jpg	0	0
test13.jpg	1	5
test14.jpg	1	6
test15.jpg	1	6
test16.jpg	1	1
test17.jpg	1	4
test18.jpg	1	4
test19.jpg	1	5
test20.jpg	0	0
test21.jpg	0	0
test22.jpg	0	0
test23.jpg	0	0
test24.jpg	1	4
test25.jpg	1	6
test26.jpg	1	7
test27.jpg	1	8
test28.jpg	0	0
test29.jpg	1	5
test30.jpg	0	0
test31.jpg	0	0
test32.jpg	0	0
test33.jpg	0	0

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

test34.jpg	0	0
test35.jpg	1	2
test36.jpg	1	6
test37.jpg	1	4
test38.jpg	1	5
test39.jpg	1	7
test40.jpg	1	2
test41.jpg	1	5
test42.jpg	1	4
test43.jpg	0	0
test44.jpg	0	0
test45.jpg	0	0
test46.jpg	0	0
test47.jpg	1	6
test48.jpg	1	4
test49.jpg	1	6
test50.jpg	1	5
test51.jpg	1	4
test52.jpg	1	1
test53.jpg	1	3
test54.jpg	1	5
test55.jpg	1	4
test56.jpg	1	4
test57.jpg	1	6
test58.jpg	1	3
test59.jpg	0	0
test60.jpg	0	0
test61.jpg	1	2
test62.jpg	1	3
test63.jpg	1	2
test64.jpg	1	3
test65.jpg	1	2
test66.jpg	0	0
test67.jpg	0	0
test68.jpg	0	0
test69.jpg	0	0
test70.jpg	0	0

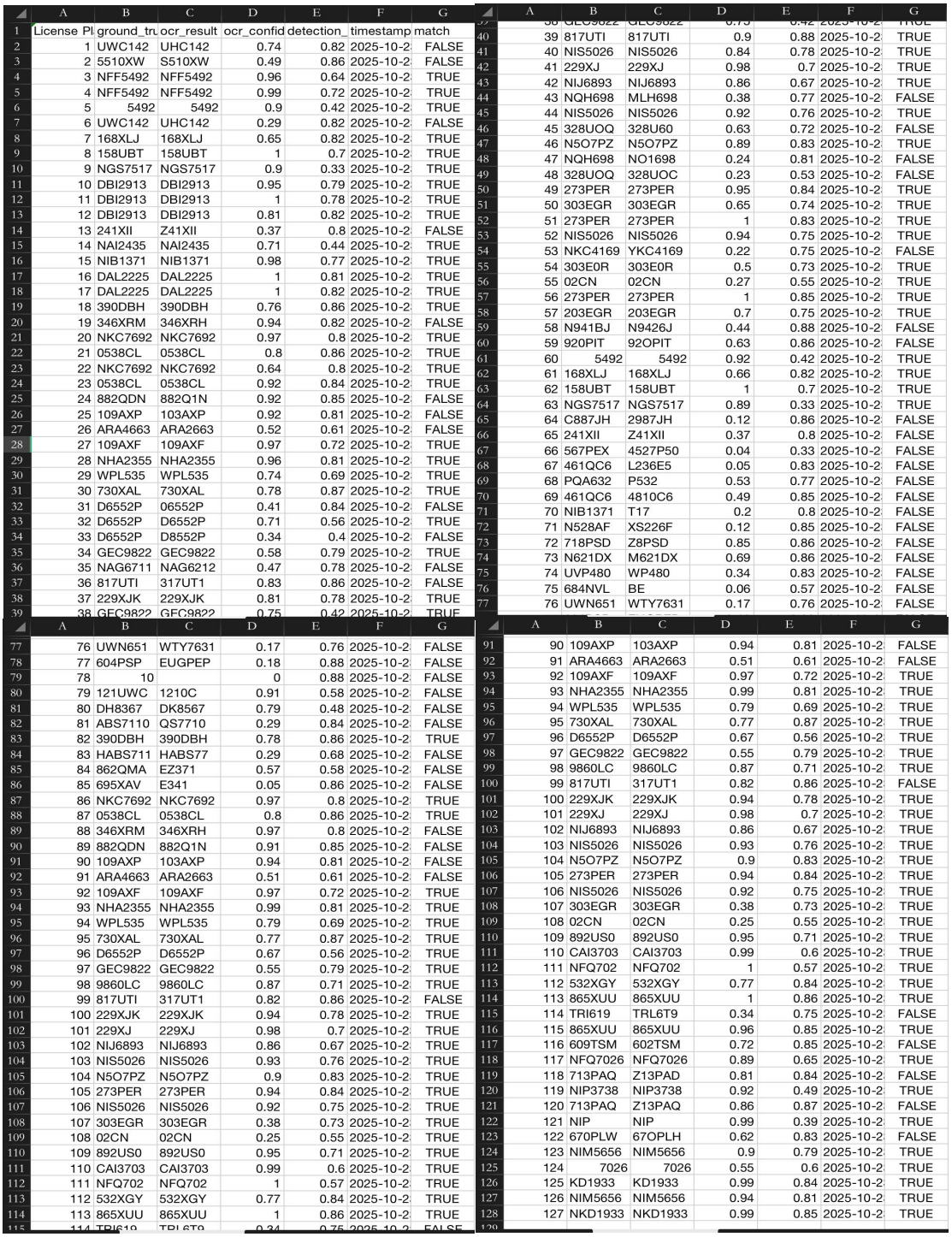
Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

test71.jpg	0	0
test72.jpg	1	6
test73.jpg	1	6
test74.jpg	1	6
test75.jpg	1	5
test76.jpg	1	6
test77.jpg	1	5
test78.jpg	1	5
test79.jpg	1	4
test80.jpg	1	4
test81.jpg	1	4
test82.jpg	1	4
test83.jpg	1	6
test84.jpg	1	4
test85.jpg	1	4
test86.jpg	1	5
test87.jpg	0	0
test88.jpg	1	6
test89.jpg	1	3
test90.jpg	1	6
test91.jpg	1	3
test92.jpg	1	5
test93.jpg	0	0
test94.jpg	0	0
test95.jpg	0	0
test96.jpg	1	3
test97.jpg	1	5
test98.jpg	1	5
test99.jpg	1	5
test100.jpg	1	4

LTO Expert :

IRIS ERVENE S. SAJOK

Signature Over Printed Name



	A	B	C	D	E	F	G
1	License Pl	ground_tru	ocr_result	ocr_confid	detection_	timestamp	match
2	1	UWC142	UHC142	0.74	0.82	2025-10-2	FALSE
3	2	5510XW	S510XW	0.49	0.86	2025-10-2	FALSE
4	3	NFF5492	NFF5492	0.96	0.64	2025-10-2	TRUE
5	4	NFF5492	NFF5492	0.99	0.72	2025-10-2	TRUE
6	5	5492	5492	0.9	0.42	2025-10-2	TRUE
7	6	UWC142	UHC142	0.29	0.82	2025-10-2	FALSE
8	7	168XLJ	168XLJ	0.65	0.82	2025-10-2	TRUE
9	8	158UBT	158UBT	1	0.7	2025-10-2	TRUE
10	9	NGS7517	NGS7517	0.9	0.33	2025-10-2	TRUE
11	10	DBI2913	DBI2913	0.95	0.79	2025-10-2	TRUE
12	11	DBI2913	DBI2913	1	0.78	2025-10-2	TRUE
13	12	DBI2913	DBI2913	0.81	0.82	2025-10-2	TRUE
14	13	241XII	Z41XII	0.37	0.8	2025-10-2	FALSE
15	14	NAI2435	NAI2435	0.71	0.44	2025-10-2	TRUE
16	15	NIB1371	NIB1371	0.98	0.77	2025-10-2	TRUE
17	16	DAL2225	DAL2225	1	0.81	2025-10-2	TRUE
18	17	DAL2225	DAL2225	1	0.82	2025-10-2	TRUE
19	18	390DBH	390DBH	0.76	0.86	2025-10-2	TRUE
20	19	346XRM	346XRH	0.94	0.82	2025-10-2	FALSE
21	20	NKC7692	NKC7692	0.97	0.8	2025-10-2	TRUE
22	21	0538CL	0538CL	0.8	0.86	2025-10-2	TRUE
23	22	NKC7692	NKC7692	0.64	0.8	2025-10-2	TRUE
24	23	0538CL	0538CL	0.92	0.84	2025-10-2	TRUE
25	24	882QDN	882Q1N	0.92	0.85	2025-10-2	FALSE
26	25	109AXP	103AXP	0.92	0.81	2025-10-2	FALSE
27	26	ARA4663	ARA2663	0.52	0.61	2025-10-2	FALSE
28	27	109AXF	109AXF	0.97	0.72	2025-10-2	TRUE
29	28	NHA2355	NHA2355	0.96	0.81	2025-10-2	TRUE
30	29	WPL535	WPL535	0.74	0.69	2025-10-2	TRUE
31	30	730XAL	730XAL	0.78	0.87	2025-10-2	TRUE
32	31	D6552P	06552P	0.41	0.84	2025-10-2	FALSE
33	32	D6552P	D6552P	0.71	0.56	2025-10-2	TRUE
34	33	D6552P	D8552P	0.34	0.4	2025-10-2	FALSE
35	34	GEC9822	GEC9822	0.58	0.79	2025-10-2	TRUE
36	35	NAG6711	NAG6212	0.47	0.78	2025-10-2	FALSE
37	36	817UTI	317UT1	0.83	0.86	2025-10-2	FALSE
38	37	229XJK	229XJK	0.81	0.78	2025-10-2	TRUE
39	38	GEC9822	GEC9822	0.75	0.42	2025-10-2	TRUE
40	39	817UTI	817UTI	0.9	0.88	2025-10-2	TRUE
41	40	NIS5026	NIS5026	0.84	0.78	2025-10-2	TRUE
42	41	229XJ	229XJ	0.98	0.7	2025-10-2	TRUE
43	42	NIJ6893	NIJ6893	0.86	0.67	2025-10-2	TRUE
44	43	NQH698	MLH698	0.38	0.77	2025-10-2	FALSE
45	44	NIS5026	NIS5026	0.92	0.76	2025-10-2	TRUE
46	45	328UOQ	328U60	0.63	0.72	2025-10-2	FALSE
47	46	N5O7PZ	N5O7PZ	0.89	0.83	2025-10-2	TRUE
48	47	NQH698	NO1698	0.24	0.81	2025-10-2	FALSE
49	48	328UOQ	328UOC	0.23	0.53	2025-10-2	FALSE
50	49	273PER	273PER	0.95	0.84	2025-10-2	TRUE
51	50	303EGR	303EGR	0.65	0.74	2025-10-2	TRUE
52	51	273PER	273PER	1	0.83	2025-10-2	TRUE
53	52	NIS5026	NIS5026	0.94	0.75	2025-10-2	TRUE
54	53	NKC4169	YKC4169	0.22	0.75	2025-10-2	FALSE
55	54	303E0R	303E0R	0.5	0.73	2025-10-2	TRUE
56	55	02CN	02CN	0.27	0.55	2025-10-2	TRUE
57	56	273PER	273PER	1	0.85	2025-10-2	TRUE
58	57	203EGR	203EGR	0.7	0.75	2025-10-2	TRUE
59	58	N941BJ	N9426J	0.44	0.88	2025-10-2	FALSE
60	59	920PIT	92OPIT	0.63	0.86	2025-10-2	FALSE
61	60	5492	5492	0.92	0.42	2025-10-2	TRUE
62	61	168XLJ	168XLJ	0.66	0.82	2025-10-2	TRUE
63	62	158UBT	158UBT	1	0.7	2025-10-2	TRUE
64	63	NGS7517	NGS7517	0.89	0.33	2025-10-2	TRUE
65	64	C887JH	2987JH	0.12	0.86	2025-10-2	FALSE
66	65	241XII	Z41XII	0.37	0.8	2025-10-2	FALSE
67	66	567PEX	4527P50	0.04	0.33	2025-10-2	FALSE
68	67	461QC6	L236E5	0.05	0.83	2025-10-2	FALSE
69	68	PQA632	P532	0.53	0.77	2025-10-2	FALSE
70	69	461QC6	4810C6	0.49	0.85	2025-10-2	FALSE
71	70	NIB1371	T17	0.2	0.8	2025-10-2	FALSE
72	71	N528AF	XS226F	0.12	0.85	2025-10-2	FALSE
73	72	718PSD	Z8PSD	0.85	0.86	2025-10-2	FALSE
74	73	N621DX	M621DX	0.69	0.86	2025-10-2	FALSE
75	74	UVP480	WP480	0.34	0.83	2025-10-2	FALSE
76	75	684NVL	BE	0.06	0.57	2025-10-2	FALSE
77	76	UWN651	WTY7631	0.17	0.76	2025-10-2	FALSE
78	77	604PSP	EUGPEP	0.18	0.88	2025-10-2	FALSE
79	78	10		0	0.88	2025-10-2	FALSE
80	79	121UWC	1210C	0.91	0.58	2025-10-2	FALSE
81	80	DH8367	DK8567	0.79	0.48	2025-10-2	FALSE
82	81	ABS7110	QS7710	0.29	0.84	2025-10-2	FALSE
83	82	390DBH	390DBH	0.78	0.86	2025-10-2	TRUE
84	83	HABS711	HABS77	0.29	0.68	2025-10-2	FALSE
85	84	862QMA	EZ371	0.57	0.58	2025-10-2	FALSE
86	85	695XAV	E341	0.05	0.86	2025-10-2	FALSE
87	86	NKC7692	NKC7692	0.97	0.8	2025-10-2	TRUE
88	87	0538CL	0538CL	0.8	0.86	2025-10-2	TRUE
89	88	346XRM	346XRH	0.97	0.8	2025-10-2	FALSE
90	89	882QDN	882Q1N	0.91	0.85	2025-10-2	FALSE
91	90	109AXP	103AXP	0.94	0.81	2025-10-2	FALSE
92	91	ARA4663	ARA2663	0.51	0.61	2025-10-2	FALSE
93	92	109AXF	109AXF	0.97	0.72	2025-10-2	TRUE
94	93	NHA2355	NHA2355	0.99	0.81	2025-10-2	TRUE
95	94	WPL535	WPL535	0.79	0.69	2025-10-2	TRUE
96	95	730XAL	730XAL	0.77	0.87	2025-10-2	TRUE
97	96	D6552P	D6552P	0.67	0.56	2025-10-2	TRUE
98	97	GEC9822	GEC9822	0.55	0.79	2025-10-2	TRUE
99	98	9860LC	9860LC	0.87	0.71	2025-10-2	TRUE
100	99	817UTI	317UT1	0.82	0.86	2025-10-2	FALSE
101	100	229XJK	229XJK	0.94	0.78	2025-10-2	TRUE
102	101	229XJ	229XJ	0.98	0.7	2025-10-2	TRUE
103	102	NIJ6893	NIJ6893	0.86	0.67	2025-10-2	TRUE
104	103	NIS5026	NIS5026	0.93	0.76	2025-10-2	TRUE
105	104	N5O7PZ	N5O7PZ	0.9	0.83	2025-10-2	TRUE
106	105	273PER	273PER	0.94	0.84	2025-10-2	TRUE
107	106	NIS5026	NIS5026	0.92	0.75	2025-10-2	TRUE
108	107	303EGR	303EGR	0.38	0.73	2025-10-2	TRUE
109	108	02CN	02CN	0.25	0.55	2025-10-2	TRUE
110	109	892US0	892US0	0.95	0.71	2025-10-2	TRUE
111	110	CAI3703	CAI3703	0.99	0.6	2025-10-2	TRUE
112	111	NFQ702	NFQ702	1	0.57	2025-10-2	TRUE
113	112	532XGY	532XGY	0.77	0.84	2025-10-2	TRUE
114	113	865XUU	865XUU	1	0.86	2025-10-2	TRUE
115	114	TRI619	TRL6T9	0.34	0.75	2025-10-2	FALSE
116	115	865XUU	865XUU	0.96	0.85	2025-10-2	TRUE
117	116	609TSM	602TSM	0.72	0.85	2025-10-2	FALSE
118	117	NFQ7026	NFQ7026	0.89	0.65	2025-10-2	TRUE
119	118	713PAQ	Z13PAD	0.81	0.84	2025-10-2	FALSE
120	119	NIP3738	NIP3738	0.92	0.49	2025-10-2	TRUE
121	120	713PAQ	Z13PAQ	0.86	0.87	2025-10-2	FALSE
122	121	NIP	NIP	0.99	0.39	2025-10-2	TRUE
123	122	670PLW	67OPLH	0.62	0.83	2025-10-2	FALSE
124	123	NIM5656	NIM5656	0.9	0.79	2025-10-2	TRUE
125	124	7026	7026	0.55	0.6	2025-10-2	TRUE
126	125	KD1933	KD1933	0.99	0.84	2025-10-2	TRUE
127	126	NIM5656	NIM5656	0.94	0.81	2025-10-2	TRUE
128	127	NKD1933	NKD1933	0.99	0.85	2025-10-2	TRUE

The figure presents screenshots of the complete Excel spreadsheet used to evaluate the performance metrics for character and full plate recognition accuracy on Google Collab.

	Images	GroundTruth	Yolov8 Model	Difference	TP	FP	FN	percentage_error
2	Image 1	5	5	0	5	0	0	0
3	Image 2	0	0	0	0	0	0	0
4	Image 3	5	5	0	5	0	0	0
5	Image 4	5	5	0	5	0	0	0
6	Image 5	0	0	0	0	0	0	0
7	Image 6	6	6	0	6	0	0	0
8	Image 7	6	6	0	6	0	0	0
9	Image 8	6	6	0	6	0	0	0
10	Image 9	6	6	0	6	0	0	0
11	Image 10	6	6	0	6	0	0	0
12	Image 11	0	0	0	0	0	0	0
13	Image 12	0	0	0	0	0	0	0
14	Image 13	4	4	0	4	0	0	0
15	Image 14	5	5	0	5	0	0	0
16	Image 15	4	4	0	4	0	0	0
17	Image 16	1	1	0	1	0	0	0
18	Image 17	4	4	0	4	0	0	0
19	Image 18	2	2	0	2	0	0	0
20	Image 19	4	4	0	4	0	0	0
21	Image 20	0	0	0	0	0	0	0
22	Image 21	0	0	0	0	0	0	0
23	Image 22	0	0	0	0	0	0	0
24	Image 23	0	0	0	0	0	0	0
25	Image 24	4	3	1	3	0	1	25
26	Image 25	4	4	0	4	0	0	0
27	Image 26	6	6	0	6	0	0	0
28	Image 27	5	5	0	5	0	0	0
29	Image 28	0	0	0	0	0	0	0
30	Image 29	5	5	0	5	0	0	0
31	Image 30	0	0	0	0	0	0	0
32	Image 31	0	0	0	0	0	0	0
33	Image 32	0	0	0	0	0	0	0
34	Image 33	0	0	0	0	0	0	0
35	Image 34	0	0	0	0	0	0	0
36	Image 35	2	2	0	2	0	0	0
37	Image 36	6	6	0	6	0	0	0
38	Image 37	4	4	0	4	0	0	0
39	Image 38	5	5	0	5	0	0	0
40	Image 39	7	6	1	6	0	1	14.28571429
41	Image 40	2	3	-1	2	1	0	50
42	Image 41	5	5	0	5	0	0	0
43	Image 42	4	4	0	4	0	0	0
44	Image 43	0	0	0	0	0	0	0
45	Image 44	0	0	0	0	0	0	0
46	Image 45	0	0	0	0	0	0	0
47	Image 46	0	0	0	0	0	0	0
48	Image 47	6	5	1	5	0	1	16.66666667
49	Image 48	4	4	0	4	0	0	0
50	Image 49	6	6	0	6	0	0	0
51	Image 50	4	4	0	4	0	0	0
51	Image 50	4	4	0	4	0	0	0
52	Image 51	4	4	0	4	0	0	0
53	Image 52	1	1	0	1	0	0	0
54	Image 53	3	3	0	3	0	0	0
55	Image 54	5	5	0	5	0	0	0
56	Image 55	5	5	0	5	0	0	0
57	Image 56	4	4	0	4	0	0	0
58	Image 57	6	5	1	5	0	1	16.66666667
59	Image 58	3	2	1	2	0	1	33.33333333
60	Image 59	0	0	0	0	0	0	0
61	Image 60	0	0	0	0	0	0	0
62	Image 61	2	2	0	2	0	0	0
63	Image 62	3	3	0	3	0	0	0
64	Image 63	2	3	-1	2	1	0	50
65	Image 64	3	4	-1	3	1	0	33.33333333
66	Image 65	2	2	0	2	0	0	0
67	Image 66	0	0	0	0	0	0	0
68	Image 67	0	0	0	0	0	0	0
69	Image 68	0	0	0	0	0	0	0
70	Image 69	0	0	0	0	0	0	0
71	Image 70	0	0	0	0	0	0	0
72	Image 71	0	0	0	0	0	0	0
73	Image 72	6	6	0	6	0	0	0
74	Image 73	6	6	0	6	0	0	0
75	Image 74	6	6	0	6	0	0	0
76	Image 75	5	5	0	5	0	0	0
77	Image 76	6	5	-1	5	0	1	16.66666667
78	Image 77	5	5	0	5	0	0	0
79	Image 78	5	5	0	5	0	0	0
80	Image 79	4	4	0	4	0	0	0
81	Image 80	4	4	0	4	0	0	0
82	Image 81	4	4	0	4	0	0	0
83	Image 82	4	4	0	4	0	0	0
84	Image 83	6	5	1	5	0	1	16.66666667
85	Image 84	4	4	0	4	0	0	0
86	Image 85	4	4	0	4	0	0	0
87	Image 86	5	5	0	5	0	0	0
88	Image 87	0	0	0	0	0	0	0
89	Image 88	6	6	0	6	0	0	0
90	Image 89	3	3	0	3	0	0	0
91	Image 90	6	7	-1	6	1	0	16.66666667
92	Image 91	3	3	0	3	0	0	0
93	Image 92	5	5	0	5	0	0	0
94	Image 93	0	0	0	0	0	0	0
95	Image 94	0	0	0	0	0	0	0
96	Image 95	0	0	0	0	0	0	0
97	Image 96	3	3	0	3	0	0	0
98	Image 97	5	5	0	5	0	0	0
99	Image 98	5	6	-1	5	1	0	20
100	Image 99	5	5	0	5	0	0	0
101	Image 100	4	4	0	4	0	0	0

Figure above shows complete Excel Table used for model validation

Formula:

1. Vehicle Detection Metrics (YOLOv8)

Let:

TP = True Positives (correctly detected motorcycles/vehicles)

FP = False Positives (incorrectly detected as motorcycles/vehicles)

FN = False Negatives (missed motorcycles/vehicles)

Precision (P):

$$Precision = \frac{TP}{TP + FP}$$

Recall (R):

$$Recall = \frac{TP}{TP + FN}$$

F1 Score:

$$F1 = 2\ x\ \frac{Precision\ x\ Recall}{Precision\ +\ Recall}$$

Mean Average Precision (mAP@0.5):

$$maAP@0.5 = \frac{AP\ for\ all\ classes\ at\ IoU\ 0.5}{Number\ of\ Classes}$$

(AP = area under the Precision-Recall curve for each class)

2. Bicycle Lane Segmentation (U-Net)

Intersection over Union (IoU):

$$IoU = \frac{|Prediction \cap GroundTruth|}{|Prediction \cup GroundTruth}$$

Dice Coefficient (F1 for segmentation):

$$Dice = \frac{2x\ |Prediction \cap GroundTruth\ |}{|Prediction\ |\ +\ |GroundTruth\ |}$$

3. License Plate Recognition (EasyOCR)

Character Recognition Rate (CRR):

$$\text{CRR} = \frac{\text{Number of correctly recognized characters}}{\text{Total number of characters}} \text{ x } 100\%$$

Plate Recognition Rate (PRR):

$$\boldsymbol{PRR = \frac{\text{Number of correctly recognized full plates}}{\text{Total number of plates}} x\ 100\%}$$

APPENDIX E

Sample Input / Output



Figure E1. Sample annotated images that will be used for model development necessary for vehicle and license plate detection and bicycle lane segmentation

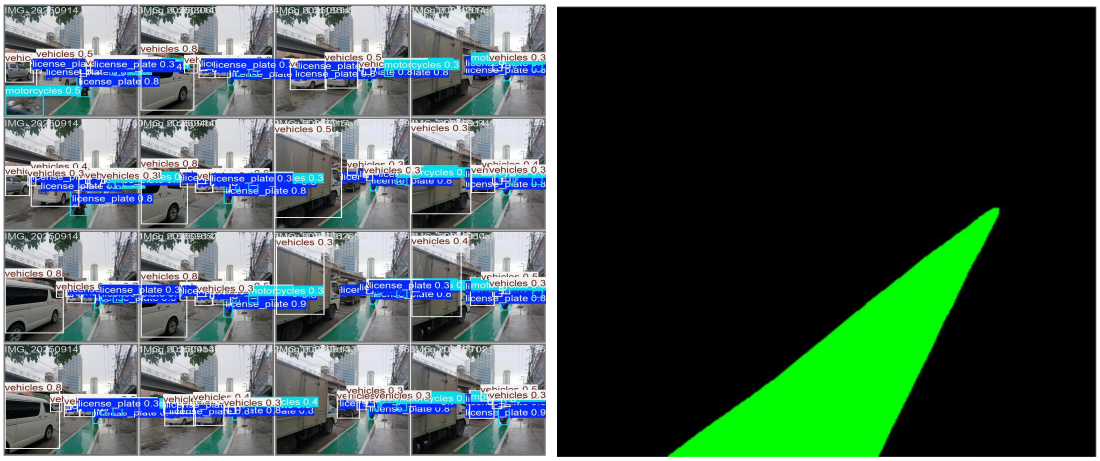


Figure E2. Sample images in training the data

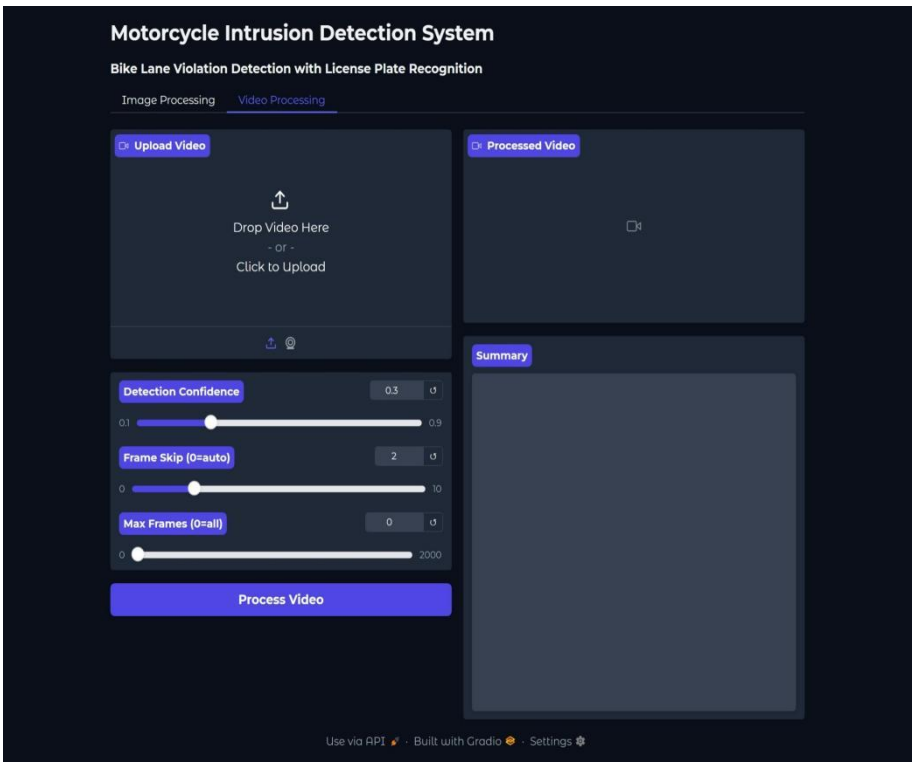


Figure E3. Sample output where the model can detect vehicles & license plate, motorcycle intrusion and recognize license plate by intergrating EasyOCR

APPENDIX F

User's Guide

Simple User Guide In Using Our Basic Model For Motorcycle Intrusion System



Upload processing

1. **Upload Video -** Drag and drop your video file into the "Upload Video" area or click to upload.
2. **Adjust Settings**
   Detection Confidence: Set the confidence level (0.1 - 1.0).
   Frame Skip: Choose how many frames to skip for processing.
   Max Frames: Specify the maximum number of frames to analyze (up to 2000).
3. **Process Video -** Click the "Process Video" button to start detection.
4. **View Results -** Check the "Processed Video" and "Summary" sections for results.

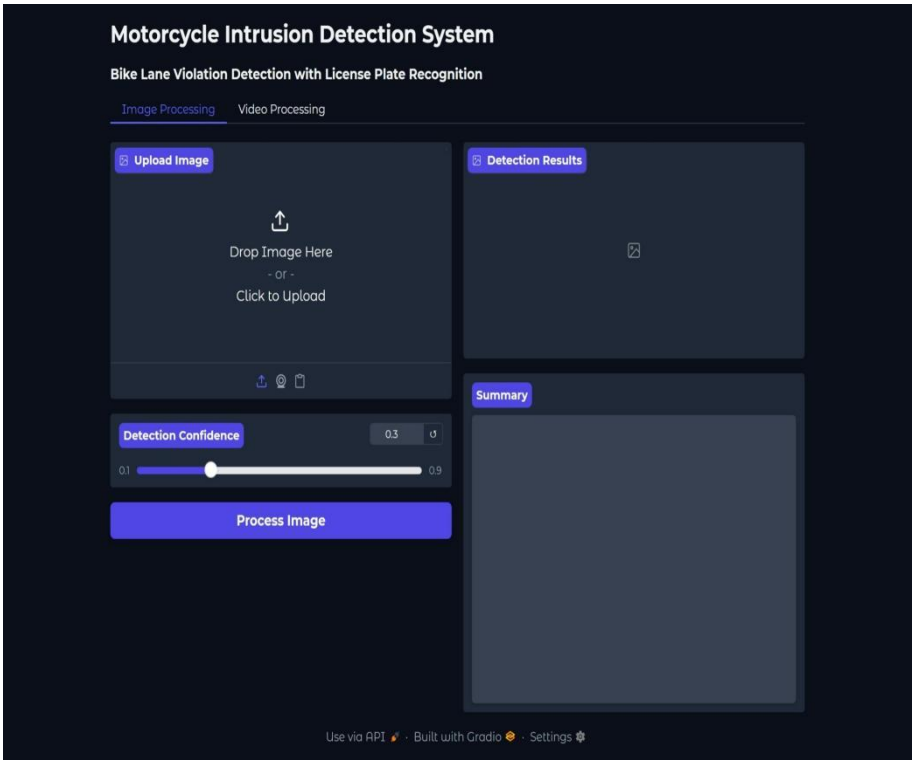


Image Processing

1. **Upload Image -** Drag and drop your image file into the "Upload Image" area or click to upload.
2. **Set Detection Confidence -** Adjust the detection confidence level (0.1 - 1.0).
3. Process Image - Click the "Process Image" button to begin detection.
4. **View Results -** Review the "Detection Results" and "Summary" sections for insights.

APPENDIX G

Matrix For Revision

MATRIX FOR REVISION

Commentors	Consultants	Comments	Page No.	Actions	Page No.
Prof. Carolina R. Joval	Statistician	The t-test result must have a no significant difference between the expert and model count for Motorcycle Intrusion Validation.	49	The threshold value for motorcycle intrusion was changed from 0.5 to 0.6 to avoid detecting far images of motorcycle on bicycle lanes. The p-value was changed to 0.594, indicating a no significant relationship.	61 & Appendix D
Prof. Wilson B. Suyat	Subject Specialist	In relation to the comment above, it was suggested to change the value of threshold in order to avoid detecting far motorcycle			49
Prof. Wilson B. Suyat	Subject Specialist	The manuscript should be revised to maintain a consistent third-person point of view by avoiding first-person pronouns such as ‘we’ and ‘I’ throughout the document.	43-63	Fixed the manuscript to consistent third-person point of view	43-63

APPENDIX H

Grammarly & Turnitin Report

CHAPTER I-3_jan_07_2026

by Grammarly

General metrics

63,393	8,889	552	35 min 33 sec	1 hr 8 min
characters	words	sentences	reading time	speaking time

Score



This text scores better than 96% of all texts checked by Grammarly

Writing Issues

102		102
Issues left	Critical	Advanced

CHAPTER 4-5_jan_07_2026

by Grammarly

General metrics

28,036	3,968	453	15 min 52 sec	30 min 31 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 96% of all texts checked by Grammarly

Writing Issues

48	1	47
Issues left	Critical	Advanced

Thesis-Mendez

ORIGINALITY REPORT

15%	11%	11%	%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	www.coursehero.com Internet Source	<1%
2	www.frontiersin.org Internet Source	<1%
3	ijeecs.iaescore.com Internet Source	<1%
4	techscience.com Internet Source	<1%
5	Alam, Tasmeer. "Bridging Computer Vision and Data Analysis: A Queryable Video Framework", Morgan State University, 2025 Publication	<1%

APPENDIX I

Curriculum Vitae



CHRIS F. BALOGO

09915193868

balogochris12@gmail.com

1871 Sitio Maligaya, San Antonio, Los Baños, Laguna

TECHNICAL SKILLS

Programming language

- Phyton
- JavaScript
- C#

Tools and Platform

- Git
- GitHub
- Visual Studio Code
- Google Colab
- Jupiter Notebook
- Pycharm

LANGUAGE

Filipino
English

ABOUT ME

I am a dedicated Computer Science student with a strong background in programming, machine learning, and computer vision. I am passionate about integrating AI models into web-based systems and continuously learning new technologies to create efficient and intelligent solutions.

PROJECTS

Lbusiness

April - May 2025

LBusiness is a website that promotes and showcases local businesses across all barangays of Los Baños, helping users discover nearby products and services while supporting local entrepreneurs.

Piyuhub

October - December 2024

PiyuHub is an app for LSPU Los Baños students to stay updated on school agendas and events, where I worked as the front-end developer ensuring a simple and user-friendly interface.

EDUCATION

2022-2026 **Bachelor of Science in Computer Science**
Laguna State Polytechnic University - Los Baños Campus

2020-2022 **Colegio De Los Baños**
Lopez Ave, Batong Malake, Los Baños, 4030 Laguna

2016-2020 **Los Baños National Highschool - Batong Malake**
Umali Subdivision, Los Baños, Laguna



DAREL T. ALCARAZ

09686708186

dareltenorioalcaraz@gmail.com

B4-L8 Cuervo Ville, Real Calamba Laguna

TECHNICAL SKILLS

Programming

Python

Web Development

HTML

CSS

Tools & Platforms

Pycharm

Google Colab

Visual Studio Code

Jupiter Notebook

LANGUAGE

Filipino

English

ABOUT ME

BS Computer Science student with foundational knowledge in programming and web development. I am currently learning AI concepts and computer vision through a hands-on thesis project. I am hardworking and dedicated, with a strong willingness to learn new technologies and continuously improve my technical skills.

PROJECTS

LBusiness

April - May 2025

LBusiness is a website that promotes and showcases local businesses across all barangays of Los Baños, helping users discover nearby products and services while supporting local entrepreneurs.

PiyuHub

October - December 2024

PiyuHub is an app for LSPU Los Baños students to stay updated on school agendas and events, where I worked as the front-end developer ensuring a simple and user-friendly interface.

EDUCATION

2021-2026 **Laguna State Polytechnic University - Los Banos, Laguna**
Bachelor of Science in Computer Science

2019-2021 **Laguna College of Business and Arts**
Science, Technology, Engineering, and Mathematics

2016-2019 **Lecheria Integrated School**
Lecheria Integrated School



RODA C. MENDEZ

09944796155

rodamendez06@gmail.com

372. Sitio Ibaba Brgy. Sto. Domingo, Bay, Laguna

TECHNICAL SKILLS

Programming

Python

Web Development

HTML

CSS

Tools & Platforms

Pycharm

Google Colab

Visual Studio Code

Jupiter Notebook

LANGUAGE

Filipino

English

ABOUT ME

A Bachelor of Science in Computer Science major in Intelligent Systems, passionate about programming and continuous learning. Adaptable and responsible, with strong problem-solving and communication skills. I strive to develop innovative and efficient solutions that create real-world impact.

PROJECTS

LBusiness

April - May 2025

LBusiness is a website that promotes and showcases local businesses across all barangays of Los Baños, helping users discover nearby products and services while supporting local entrepreneurs.

PiyuHub

October - December 2024

PiyuHub is an app for LSPU Los Baños students to stay updated on school agendas and events, where I worked as the front-end developer ensuring a simple and user-friendly interface.

EDUCATION

2021-2026 **Laguna State Polytechnic University - Los Banos, Laguna**
Bachelor of Science in Computer Science
Consistent Dean's Lister

2019-2021 **Los Banos Senior High School**
Science, Technology, Engineering, and Mathematics
Consistent Honor

2016-2019 **Nicholas L. Galvez Memorial Integrated National High School - Bay, Laguna**
Nicholas L. Galvez Memorial Integrated National High School
Consistent Honor

APPENDIX J

Publishable Paper (IEEE format)

UTILIZATION OF DEEP LEARNING: OPTIMIZED YOLOV8 FOR DETECTION OF MOTORCYCLE INTRUSION AND LICENSE PLATE RECOGNITION ON BICYCLE LANES

Darel T. Alcaraz, Chris F. Balogo, Roda C. Mendez, Jefferson L. Lerios
College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

Article Info

Keywords:
You Only Look Once Version 8 (YOLOv8)
Motorcycle
EasyOCR,
Bicycle Lanes
U-Net
ResNet34
Detection
Annotation
Recognition
Deep Learning

ABSTRACT

The study utilized a deep learning algorithm, specifically optimizing the YOLOv8 to detect motorcycle intrusions and recognize license plates using EasyOCR on bicycle lanes. As motorcycle usage rises in urban areas, the need for effective monitoring systems becomes critical for protecting cyclists and pedestrians. The research involves collecting and annotating a dataset of 1,832 images using Roboflow, leading to the development of two models: YOLOv8 for vehicle and license plate detection and U-Net with ResNet34 for bicycle lane segmentation. Model performance was assessed using metrics like Precision, Recall, F1 Score, and Mean Average Precision (mAP) for detection, and Dice coefficient and Intersection over Union (IoU) for segmentation accuracy. EasyOCR was also integrated for license plate recognition, evaluated through Character Recognition Rate (CRR) and Plate Recognition Rate (PRR). Results indicated that the YOLOv8 model achieved 88% precision and 91% mAP while the segmentation model had an IoU of approximately 0.975, and EasyOCR demonstrated a character accuracy of 83.82%. The YOLOv8 model detected motorcycles on bicycle lanes with high accuracy, resulting in a precision of 94.74%, recall of 87.54%, and an F1-score of 91%. Further, statistical test results showed no significant relationship between the expert count and the model count, with a p-value of 0.594. These results show that the model's performance is closely aligned with the experts' assessment, supporting the reliability of YOLOv8 model in detection of motorcycle intrusion on bicycle lanes.

Corresponding Author:

Darel T. Alcaraz
College of Computer Studies
Laguna State Polytechnic University
Malinta, Los Baños, Laguna, Philippines, 4030
Email: dareltenorioalcaraz@gmail.com

1. INTRODUCTION

The increasing use of motorcycles on city roads has raised serious concerns regarding traffic violations, particularly the unauthorized entry of motorcycles into bicycle lanes, endangering pedestrians and cyclists. The World Health Organization (2023) reports that road traffic injuries lead to approximately 1.19 million deaths annually, with vulnerable road users—pedestrians (21%), cyclists (5%), and motorcyclists (30%)—making up a significant portion of these fatalities [11]. Additionally, the National Highway Traffic Safety Administration (NHTSA) noted that in

2023, 1,166 bicyclists were killed and about 49,489 injured in the United States, illustrating the impact of inadequate enforcement in protecting non-motorized road users [5].

In the Philippines, motorcycle accidents represent a growing challenge, with around 8.5 million registered motorcycles and over 26,599 reported accidents in 2022, a number that increased by 17.3% in 2023, translating to approximately 78 accidents per day [2][4][7]. This highlights the urgent need for stricter enforcement of traffic laws to protect vulnerable users. The concept of smart cities, which integrate technology to improve urban living, can enhance traffic safety and efficiency by using advanced monitoring systems to detect violations such as motorcycles entering bicycle lanes [1].

Recent advancements in deep learning and computer vision, particularly in object detection, offer promising solutions for these challenges. Frameworks like YOLOv8 have shown significant improvements in accuracy and speed, achieving up to 86% accuracy in license plate recognition when combined with EasyOCR [6][9]. Automatic License Plate Recognition (ALPR) systems are crucial for enforcing traffic laws and addressing issues like unauthorized access and vehicle theft [3]. By integrating modern ALPR technologies with existing traffic monitoring systems, cities can more effectively manage motorcycle intrusions into bicycle lanes, contributing to safer urban environments [8][10].

Objectives of the Study

The foremost objective of the study is utilizing deep learning to develop a model by optimizing YOLOv8 to detect motorcycle intrusion and recognize license plate entering the designated bicycle lanes. Further, the study aims to accomplish the following:

1. To collect and annotate images of motorcycles, bicycle lanes, Philippine motorcycle license plates, and other road vehicles on Roboflow, apply preprocessing techniques and prepare segmented bicycle lane data as region of interest(ROI).

2. To develop and train a detection model that identifies motorcycle intrusion using YOLOv8 and U-Net with ResNet34 for segmentation, and to apply EasyOCR to extract and recognize the alphanumeric characters found on Philippine motorcycle license plates.

3. To evaluate the performance of the optimized YOLOv8 and U-Net models in detecting motorcycle intrusions, recognizing license plates, and accurately segmenting bicycle lanes. Analyzing key performance metrics such as Precision, Recall, F1 Score, and Mean Average Precision (mAP@0.5) for vehicle detection, alongside the Dice coefficient and Intersection over Union (IoU) for bicycle lane segmentation. Additionally, the performance of EasyOCR will be assessed through Character Recognition Rate (CRR) and Plate Recognition Rate (PRR). Lastly, to validate model's performance through statistical method.

2. METHODOLOGY

In this figure, it outlines the complete process of License Plate Recognition in Motorcycle Intrusion on Motorcycle Detection on Bicycle lanes. It begins with collecting and annotating images of motorcycles, license plates, and lane boundaries. During Pre-processing, images are resized, normalized, augmented – to enhance segmentation and detection. The core Process phase uses YOLOv8s for vehicle and license plate detection, U-Net with ResNet34 for instance segmentation of Bicycle Lane, and EasyOCR for license plate recognition. The Output phase presents the detection of motorcycle intrusion on bicycle lanes and model performance metrics from utilizing a deep learning algorithm.

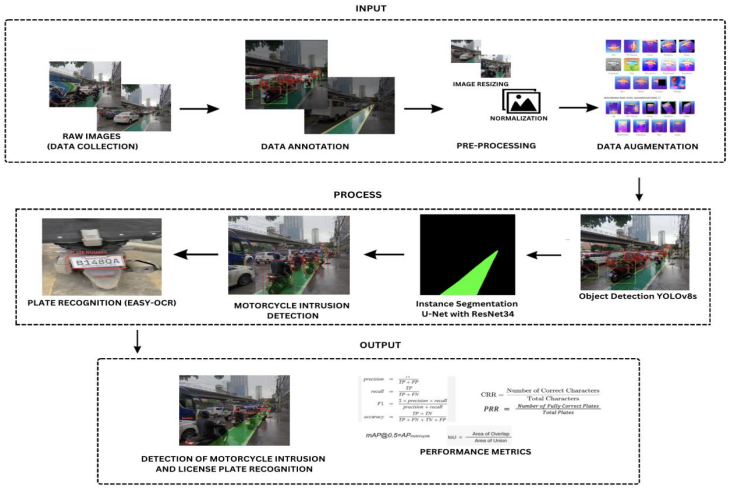


Figure 1. Conceptual Framework

3.1 DATA COLLECTION INSTRUMENT

Datasets were collected and managed using Roboflow, a comprehensive computer vision platform that facilitates data collection, annotation, and pre-processing. A dataset of 1,691 images of motorcycles, Philippine license plates, bicycle lanes, and other vehicles was collected and annotated using the Roboflow platform. Pre-processing involved standardizing resolution, color, and applying data augmentation techniques to improve data quality.

3.2 MODEL SELECTION

This study presents a systematic, multi-stage deep learning framework for detecting motorcycle intrusions and recognizing license plates in bicycle lanes. The process begins with the YOLOv8 model for initial object detection, which feeds into the U-Net + ResNet34 segmentation model to define regions of interest (ROI). The refined information is then passed to the EasyOCR module for accurate text extraction.

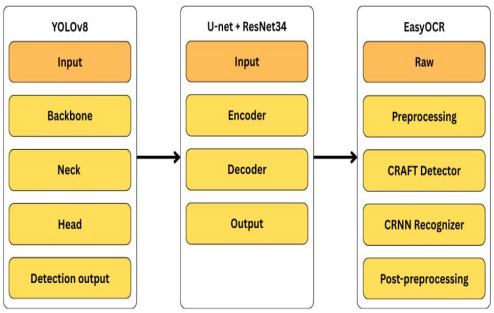


Figure 2. Model Architecture

3. RESULTS AND DISCUSSION

Object Detection

The YOLOv8 model's learning process was significantly optimized using specific hyperparameters to boost detection accuracy and reliability. Key adjustments included training for 50 epochs with a batch size of 16 and increasing the input image size to 960 pixels for improved feature extraction. The AdamW optimizer was implemented, replacing the standard optimizer, with the learning rate dynamically adjusted via a cosine schedule starting at 0.001. To prevent the model from overfitting, weight decay was reduced to 0.0001, and powerful MixUp and Copy-Paste data augmentations were heavily utilized (rate of 0.5). Further, the results underscore the high efficiency of the optimized YOLOv8 model for monitoring traffic and improving urban safety systems.



Figure 3. Trained Sample Result Using YOLOv8

Instance Segmentation

In this section, researchers focused on the bicycle lane segmentation model, which aims to accurately identify and delineate bicycle lanes in various urban environments. The training process involved a carefully curated dataset containing diverse examples of bicycle lanes under different conditions, ensuring that the model could generalize well to real-world scenarios.

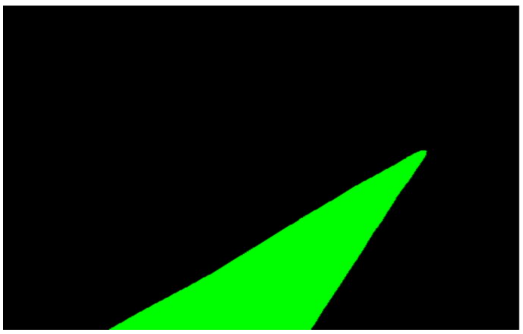


Figure 4. Trained Sample Result on Bicycle Lane Segmentation

License Plate Recognition

License plate recognition was executed using EasyOCR within a Google Colab environment, utilizing a selective sample dataset to ensure a variety of plate images were analyzed. To achieve high recognition accuracy, two essential pre-processing techniques were applied: denoising and simple grayscale conversion. The denoising step successfully reduced image noise, making the characters more distinct, while grayscale conversion enhanced the image contrast and overall clarity. This combined pre-processing approach significantly improved EasyOCR's performance, leading to higher rates of both character and complete plate recognition.

Motorcycle Intrusion on Bicycle Lane

The integrated model successfully detects and labels motorcycle intrusions in bicycle lanes. The YOLOv8 model identifies both motorcycles, marking those that enter the lane as "intrusion," with threshold value of 0.6. The U-Net segmentation model supports this by creating detailed masks that clearly outline the critical lane areas. Custom code automatically designates the intruding motorcycles as

violators, and a unique ID is given to each detected object for tracking. This combined approach allows the model to accurately spot violations and recognize license plates, distinguishing compliant vehicles from intruders.



Figure 5. Result for Detection of Motorcycle Intrusion and LPR

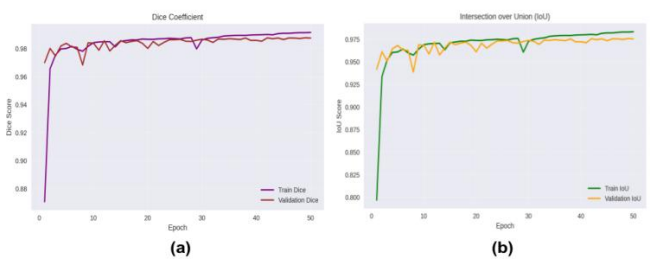
Model Performance Metrics of YOLOv8, U-Net with ResNet34, and EasyOCR

The implementation of the YOLOv8s model for motorcycle intrusion detection and license plate recognition has yielded promising results, establishing a foundation for effective monitoring. The model demonstrated high performance across key metrics, including precision, recall, and mean Average Precision (mAP), as detailed in the Table below.

Table 1. Detection of Vehicle and License Plate Training Results (YOLOV8s)

Metric	Baseline	Optimized	Improvement	Changes
Precision	0.89	0.88	-0.0041	-0.46%
Recall	0.86	0.90	+0.0427	+4.97%
F1-Score	0.87	0.89	+0.0194	+2.22%
mAP@50	0.91	0.93	+0.0266	+2.94%
mAP@50-95	0.63	0.64	+0.0138	+2.21%

The model's performance in bicycle lane segmentation was evaluated using the Dice Coefficient and Intersection over Union (IoU), which quantify the accuracy and overlap between the predicted and actual lane regions. Analysis starts by reviewing the training outcomes to confirm the model's effectiveness, followed by a detailed examination of IoU as the key metric for segmentation accuracy.



(a) (b)

Figure 6. Results in (a) Dice Coefficient and (b) Intersection over Union (IoU)

The bicycle lane segmentation model achieved exceptionally high performance, with the Dice coefficient stabilizing near 0.98 and IoU scores around 0.975 for both training and validation data. These metrics confirm the model's robustness and reliability, demonstrating high accuracy and strong agreement between its predicted segments and the actual ground truth areas.

The EasyOCR evaluation for license plate recognition, based on a dataset of 128 rows yielding 126 valid samples from 65 processed images, shows mixed performance but significant potential. High OCR confidence scores (above 0.9) generally correlate with accurate matches, but lower scores and detection challenges—often related to image quality—highlight areas needing improvement, particularly in preprocessing and detection algorithms to reduce false positives. The model achieved an exact match accuracy of 61.11% (77 out of 126), meaning more than half of the plates were fully recognized, and a strong character accuracy of 83.82%, confirming that most individual characters were correctly identified. Therefore, continuous optimization is essential to enhance the model's reliability across diverse scenarios.

Model Validation of Vehicle and License Plate Detection, Bicycle Lane Segmentation, and Motorcycle Intrusion Wilcoxon Test

Figure 7 displays the training and validation loss metrics for a machine learning model, with all metrics (bounding box, classification, and derived loss) consistently decreasing over the epochs. This continuous decline in both the top row (training losses) and the bottom row (validation losses) confirms that the model is effectively learning, improving its prediction accuracy over time, and generalizing well to new, unseen data.

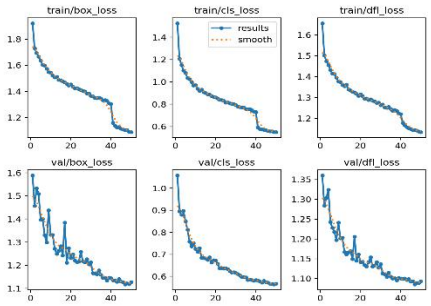


Figure 7. Validation Losses and Learning Rate During Training

The training loss decreased consistently, indicating effective learning, while the validation loss remained low, suggesting that the model generalizes well to unseen data, as illustrated at Figure 8.

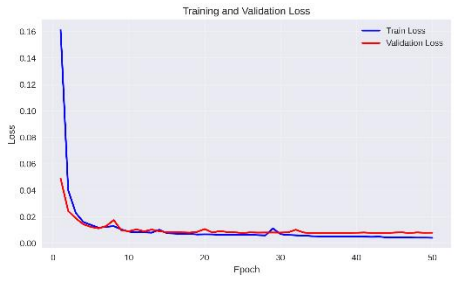


Figure 8. Training and Validation Loss

Table 2 presents the result of the paired t-test with statistic of 45.5a, for the degree of difference between two related groups. The p-value of 0.594 suggest that the difference observed is not statistically significant, implying no meaningful difference between the groups. With 100 pairs of observations, the mean difference is 0.500, showed one group has average of 0.5 units higher than the other. The standard error of the mean difference is 0.0348, reflect the reliability of this estimate. Overall, the data indicated that the two groups are effectively similar, and any observed differences are likely due to random variation rather than a substantive effect.

Table 2. Statistical Test Result

Metrics	Value
Statistic	45.5
p-value	0.594
Number of Pairs (N)	100
Mean Difference	0.500
Standard Error Difference	0.0348

4. CONCLUSION AND RECOMMENDATIONS

In conclusion, this study developed an optimized YOLOv8 model for detecting motorcycle intrusions and recognizing license plates on bicycle lanes, enhancing cyclist and pedestrian safety. With a dataset of 1,832 annotated images, YOLOv8 achieved 88% precision and 91% mAP, while the U-Net with ResNet34 model achieved an IoU of 0.975. EasyOCR integration resulted in 83.82%-character accuracy for license plate recognition.

The validation process demonstrated that the Yolov8 model performed effectively in detecting motorcycles in bicycle lanes, as evidenced by the analysis using 100 images with a precision of 94.74%, recall of 87.54%, and an F1-score of 91%. The performance metrics affirmed the model's strength and reliability. Furthermore, results of the Wilcoxon signed-rank test confirmed a no significant difference between the expert and model assessment with p-value of <0.594, indicate that the proposed model is reliable and produce results comparable to expert evaluation.

To enhance model performance for motorcycle intrusion and license plate recognition, future efforts should prioritize diversifying the dataset by incorporating various road types, weather conditions, and camera angles to improve robustness against occlusions and lighting shifts. Upgrading to state-of-the-art algorithms, such as the latest YOLO versions, will optimize detection speed and accuracy, while expanding object classes to include cars, pedestrians, and bicycles will reduce classification errors. Finally, training a dedicated recognition model on a larger variety of plate styles and viewpoints is essential for ensuring clear character visibility and reliable identification in complex environments.

REFERENCES

[1] Babu, A. M., Akhil, B., & Pochampally, N. K. (2023). Smart Cities and Intelligent Transport Systems. International Journal of Applied and Structural Mechanics, 31,22https://doi.org/10.55529/ijasm.31.22.25

[2] Daily Tribune. (2024, July 16). Daily accidents and elusive progress. The Philippine Star.

[3] Indira, K., Mohan, K. V., & Nikhilashwary, T. (2019). Automatic license plate recognition. In Recent Trends in Signal and Image Processing: ISSIP 2017 (pp. 67-77). Springer Singapore.

[4] Metropolitan Manila Development Authority. (2024, April 9). MMDA logs 17.3 % increase in motorcycle-related crashes in 2023 [News release]. Inquirer.net. Inquirer.net

[5] National Highway Traffic Safety Administration. (2023). Bicycle safety. Retrieved from [https://www.trafficsafetymarketing.gov/safety-topics/bicycle-safety]

[6] Patil, S. S., Patil, S. H., Pawar, A. M., Bewoor, M. S., Kadam, A. K., Patkar, U. C., ... & Sharma, S. (2023, July). Vehicle number plate detection using yolov8 and easyocr. In 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT) (pp. 1-4). IEEE.

[7] Philippine Statistics Authority. (2024). “As of 2023, official data indicate that there are 8.5 million registered motorcycles in the country.” Senate of the Philippines+2Philstar+2

[8] Salsabila, N. (2024). Enhancing Automated Vehicle License Plate Recognition with YOLOv8 and EasyOCR. Journal of Information Systems and Informatics, 6(3).

[9] Sohan, M., Sai Ram, T., & Rami Reddy, C. V. (2024). A review on yolov8 and its advancements. In International Conference on Data Intelligence and Cognitive Informatics (pp. 529-545). Springer, Singapore.

[10] Vedhaviyassh, D. R., Sudhan, R., Saranya, G., Safa, M., & Arun, D. (2022, December). Comparative analysis of easyocr and tesseractocr for automatic

license plate recognition using deep learning algorithm. In 2022 6th International Conference on Electronics, Communication and Aerospace Technology(pp. 966-971). IEEE.

[11] World Health Organization. (2023). Road traffic injuries. Retrieved from https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries.